



ESP Current Predictions and Troubleshooting Guidelines

BP0757

Final Revision 2

25th November 2004

Prepared by:	Reviewed by:	Accepted by
Mark Denny (BP) ESP Engineer Wytch Farm Alasdair Pirie (Helix-RDS) Petroleum Engineer	John Davies (BP) Adam Downie (Helix-RDS) Mike Hefley (BP) Yvonne Hepburn (BP)	Geoff Weighill (BP)

COVER / APPROVAL / AMENDMENT SHEET

Document No: BP0757

Document Type: FINAL REPORT

Title: ESP Current Predictions and Troubleshooting

2	25-Nov-2004	MS Word	FINAL 2 (Format Changes)	M Denny / A. Pirie		G Weighill
1	23-Sep-2004	MS Word	FINAL	M Denny / A. Pirie	J Davies, A Downie M Hefley, Y Hepburn,	G Weighill
0	July 2003	MS Word	DRAFT	M Denny		
Rev	Date	Format	Description	Originator	Reviewed	Accepted

DISTRIBUTION LIST:

Name

Geoff Weighill

Copy No.

A4 Ring Binder (001)
A5 Wire Bound (002-010)
MS Word File
PDF File

CONTENTS

1.0 INTRODUCTION	5
2.0 ESP SYSTEMS OVERVIEW	6
2.1 <i>ESP SYSTEM DESIGN</i>	6
2.1.1 ESP Optimisation Lifecycle	7
2.1.2 General Design Considerations	7
2.2 <i>ESP SIZING</i>	8
2.2.1 Initial design	8
2.2.2 Optimising the design	9
2.2.3 Final checks	9
2.3 <i>ESP COMMISSIONING AND PUMP STARTING</i>	10
2.4 <i>SAMPLE ESP SUMMARY SHEET</i>	12
2.5 <i>ESP OPERATIONS</i>	13
2.5.1 Pump starting	13
3.0 CURRENT LOADING PREDICTIONS IN ESP SYSTEMS	14
3.1 <i>SECTION INTRODUCTION</i>	14
3.2 <i>EXAMPLE 1 - NORMAL OPERATION</i>	15
3.3 <i>EXAMPLE 2 - UNDERSIZED PUMP</i>	23
3.4 <i>EXAMPLE 3 - OVERSIZED PUMP</i>	27
3.5 <i>EXAMPLE 4 - VARIABLE SPEED SYSTEM</i>	31
4.0 TROUBLESHOOTING	39
4.1 <i>NORMAL OPERATION - GENERAL</i>	39
4.1.1 Power Cable Losses	39
4.2 <i>UNDERSIZED PUMP</i>	40
4.2.1 Pump Operating Conditions – Upthrust	40
4.2.2 Overloaded Motor	40
4.2.3 Corrective Action Possible	41
4.2.4 Considerations when Inflow Performance Uncertainties Exist	43
4.3 <i>OVERSIZED PUMP</i>	44
4.3.1 Increased Power and Current Requirements	44
4.3.2 Transformers	45
4.3.3 Pump Operating Conditions - Downthrust	46
4.3.4 Well Pump Off and Gas Locking	46
4.3.5 Underload Protection	48
4.3.6 Deadheading	48
4.4 <i>VARIABLE SPEED DRIVE SYSTEMS</i>	51
4.4.1 Effects of Variable Speed Systems on Pumps	51
4.4.2 Affinity Laws	51
4.4.3 Effects of Variable Speed Systems on Motors	52
4.4.4 Summary of Frequency Change Effects	53
4.4.5 Analysis of Frequency Change Effects on Well Performance	53
4.4.6 Potential VSD Benefits – Changing Well Conditions	54
4.4.7 Additional VSD Benefits and Considerations	55

INTRODUCTION

This document is intended as a guide to Petroleum Engineers and Well Service Engineers to help them understand electrical current loading on electrical submersible pump (ESP) systems and use this to optimise performance, troubleshoot problems and design suitable replacement units for workovers. The document fills a gap in the industry documentation and is not intended to cover all aspects of ESP system design covered in vendor manuals and artificial lift publications.

The document focuses on the electrical system and uses worked examples to determine motor current using pump and motor charts and basic field data. The section on troubleshooting examines how to apply the information from the worked examples to various operational problems.

The worked examples make reference to various equipment specifications; pump and protector curves, motor curves and other miscellaneous information. It is recognised that these data will not apply to every ESP installation, however the examples are structured so that data can be substituted. The worked examples also all begin with a list of well data that should be available to the wellsite engineer.

The main application for this document is BP-TNK operations in Russia, hence some references are made to Russian equipment, e.g. Borets motor curves. However the structure of the examples is such that any vendor specifications and curves can be applied to the principles demonstrated. The majority of the BP Russian ESP operations do not have downhole pressure gauges, therefore pressure gauge information is not covered within this document.

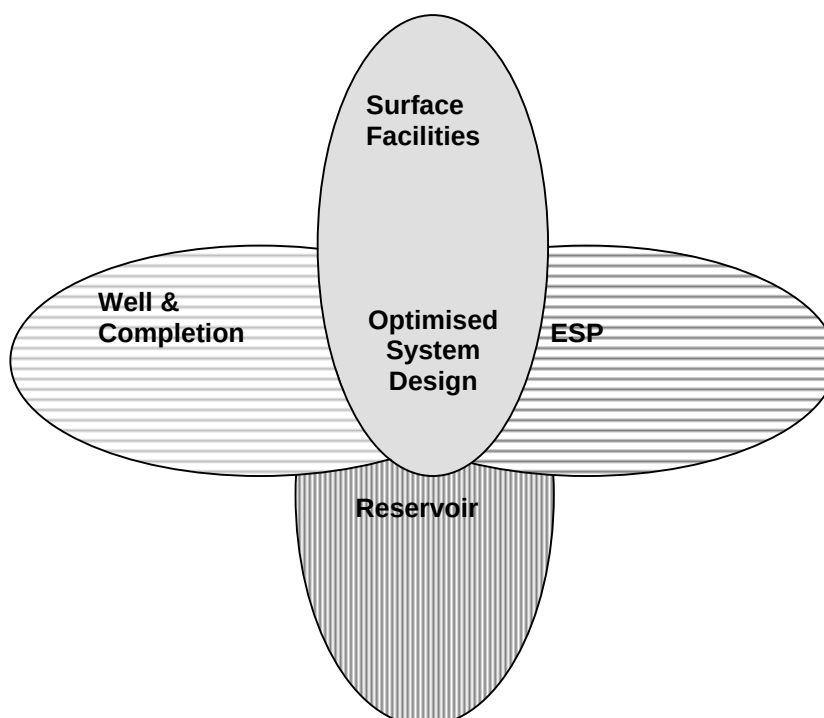
1.0 ESP SYSTEMS OVERVIEW

1.1 ESP SYSTEM DESIGN

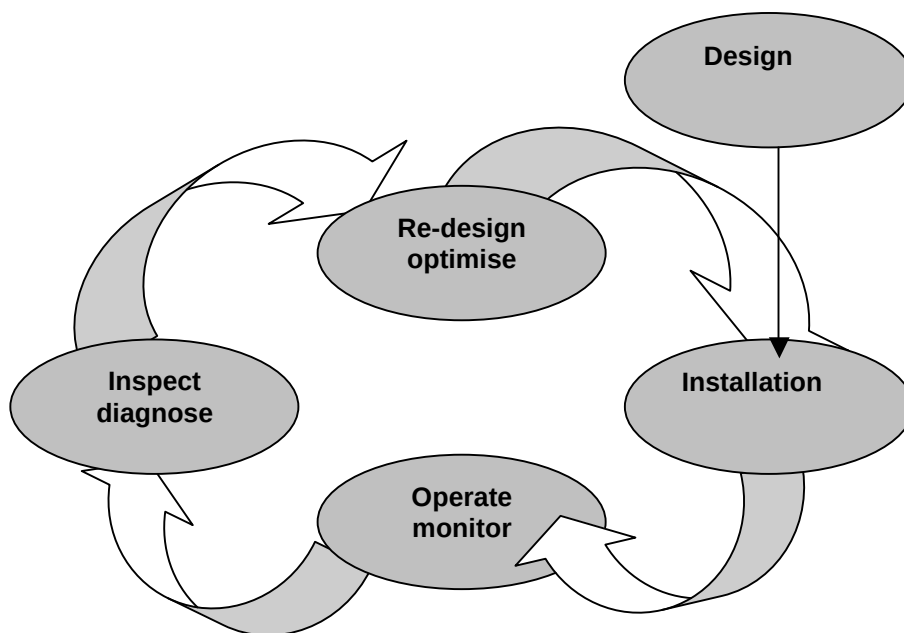
An effective ESP artificial lift system cannot be designed in isolation; all components of the production system are mutually influential. Unless the interactions between, and limitations of, each component part of the whole are considered, the system will at best be sub-optimal and at worst may fail.

By way of example, the size of the tubing will influence the ESP performance. The pressure losses and potential erosion in undersize tubing is well understood. However if the tubing is too large the flow velocity may not be sufficiently above the slip velocity resulting in water hold-up. Pump sizing software does not account for hold-up and in an extreme case the pump could be required to support a column of water in a dry well, the pump will be understaged, operate in severe downthrust, suffer motor cooling problems, fail to produce the rates anticipated and perhaps fail prematurely.

Similarly motors which are not matched to the surface electrical systems may be limited in voltage or current and hence frequency of operation (in variable frequency systems) potentially limiting the flexibility to tune the pump to changing reservoir conditions.



1.1.1 ESP Optimisation Lifecycle



1.1.2 General Design Considerations

In general the larger the diameter of the pump the more efficient it will be, however if the well performance is uncertain consideration should be also given to the shape of the pump curve.

Highly efficient pumps often suffer from poor performance outside a narrow recommended operating range. For new wells where the likely inflow performance is uncertain, a pump with a slightly lower peak efficiency but a flat efficiency profile across a broad allowable operating range would be preferable.

Generally the larger the diameter of the motor the more efficient it will be, and, for the same power output, shorter. This will reduce the susceptibility to damage in transit and whilst RIH.

Motor winding insulation degrades at a rate proportional to it's operating temperature, operating motors at less than 100% full load current load factors will reduce the internal winding temperature and increase its operational life. If workover costs are high it is generally cost effective to over-specify the motor.

Because of thrust issues, the same considerations do not apply to pumps. If in doubt about well performance the pump should be specified slightly smaller than the expected well rate, as the well can choked into the range of the pump, whereas if the pump is too large no action can be taken to reduce thrust loading.

In high angle wells labyrinth seals should be avoided, whereas in vertical wells bag seals can be the less reliable option particularly if the well fluids are aggressive to elastomers. Tandem seals can be run to increase reliability.

In high GOR wells, gas separators and gas handlers can be deployed either individually or in series. If the well is packerless annulus venting options can also reduce the free gas percentage at the pump intake. Mixed flow stage designs are less susceptible to gas locking than equivalent radial stages and should be sized to operate at total fluid (liquid and gas) flow rates at the upper end of their recommended flow rate range where they suffer less head degradation due to gas.

In wells producing solids mixed flow stages are preferred, and the vendor should be consulted regarding abrasion resistant pump bushings and impellor coatings. Potentially more serious than long-term erosion of the pump is the potential for plugging and sticking (refer to operations section for procedure when attempting to free a stuck ESP).

1.2 ESP SIZING

1.2.1 Initial design

- Choose a production rate based on well PI, reservoir pressure and bubble point (design rate may be determined by other factors such as erosion limits, reservoir management issues etc, however in this section it is assumed that the design basis is bubble point).
- Determine bottom hole flowing pressure at perforation depth.
- Set the pump depth such that the pressure at the pump intake is at the limit of free gas handling capacity of the pump.
- Check the head requirement of the pump and choose the stage count and frequency such that the pump is operating at its best efficiency point. The frequency should ideally be as close as possible to 60hz to get the maximum power from the motor (below 60 Hz the power available is: rated hp x [running frequency / 60], and above 60hz motor current is derated to reduce heating effect).
- The expected changes in water cut, reservoir pressure and other production constraints during the entire ESP installation life must be considered during the design process.
- Check the expected motor loading.
- If the load is too high the initial production rate is too high and should be dropped slightly.
- If the load is low, there is potential to increase the design flowrate, however if the flowrate takes the flowing pressure below bubble point a smaller motor may be possible.

1.2.2 Optimising the design

- Iterate through the loop again.
- Note if the rate is decreased the bottom hole flowing pressure will increase, so the pump can be raised in the well to keep the same intake pressure and vice versa.
- The final optimal solution is when:
 - o The pump is operating at its free gas limit for pump intake pressure (minimum cable voltage losses and tubing friction losses, maximum gas lift effect above and below the pump).
 - o The motor is fully loaded (motors are increasingly inefficient below 65% of full load, and larger motors are more expensive).
 - o The pump flowrate is as close as possible to the BEP of the pump (minimum conversion losses, shaft hp to hydraulic hp).
 - o The motor is at or is close to 60hz (no derating of the motor, for heating or running less than 60hz).

1.2.3 Final checks

- Finally double check that at the design operating point the following parameters are acceptable:
 - o Cable heating
 - o Cable voltage drop
 - o Available surface voltage
 - o Motor and pump shaft torque
 - o Pump housing differential burst pressure
 - o Pump position in the well with regard to any wellpath doglegs.
- Make adjustments as necessary to address these limits.

1.3 ESP COMMISSIONING AND PUMP STARTING

- An ESP system encompasses, electrical, mechanical, instrument, process, well and reservoir engineering disciplines.
- The commissioning engineer should be familiar with all aspects of these disciplines in as much as they affect the performance and operation of the system.
- Where this is not possible a single lead engineer **MUST** be delegated for the commissioning process and all decision making regarding the operation of the ESP, until steady state flow has been established and the well handed over to operations.
- To allow timely decision making, large committees of individual discipline engineers should be avoided, particularly when starting high hp systems (>500 hp) where irreversible damage can be caused by a relatively short period (30-35 minutes) of incorrect operation.
- A well handover package from the rig should document all operations carried out and the final condition of the well.
- Following workover the reinstatement of process electrical and instrument services should be witnessed along with any pressure, insulation and function tests.
- It is recommended that downhole gauges be installed in ESP wells ideally providing tubing and annulus pressure, ESP suction and discharge pressure and motor temperature. Where installed this data must be available and on line prior to startup. If downhole monitoring is not available then startup must be monitored using motor current and frequency, well head pressure and surface flowrate.

The final line up and start of the pump should be the responsibility of the lead engineer. It is recommended that a checklist be completed including:

- Tree and annulus valve status.
- Process valve alignment.
- Confirmation of flowline pressure test.
- Confirmation of instrument and control function tests.
- Electrical insulation and motor balance readings.
- Transformer tap changer settings.
- Normal pressure signature of SSSV.
- VSD setup parameters.
- Motor controller setup.

When starting the pump, the theoretical head/flow curve for the pump should be available with the head axis plotted in pressure units calculated using the s.g. of the completion fluid. Where a VSD is employed multifrequency head curves should be available covering the range of operating frequencies expected. Comparison of the head curve with the observed downhole pressures allows rapid confirmation of correct pump rotation.

Where any doubt exists the pump should be reversed and readings of:

- Pump intake pressure.
- Pump discharge pressure.
- Motor current.
- Wellhead pressure.
- Wellhead temperature.

should be compared at the same frequency of operation to determine the correct rotation. The correct rotation will be obvious given two or more of these readings.

Repeated changes of rotation should not be necessary and must be avoided as they can be extremely detrimental to the longevity of the pump. (Most high load thrust bearing systems are designed to operate in a particular direction).

In general the larger and more axial the pump stage design, the greater the difference between correct and incorrect rotation. Smaller radial stage pumps can draw similar loadings in each direction but in general fail to establish much differential pressure. In assessing pump performance consideration should be given other factors including the possibility of incorrect gauge calibration, pump sticking, pump plugging, incorrect process valve alignment etc.

When handing the well over to operations a detailed summary of the installed system and its operating constraints should be compiled including a detailed description of; the installed ESP, surface equipment, transformer ratios, well and reservoir data and most importantly, comments detailing critical start-up or operating issues.

1.4 Sample ESP Summary Sheet

Recent Well Data						
Well	PI	BHCIP	W. Cut	Oil Rate	Water Rate	Run-Life
IF-19	7.90	2158	86.2%	1388	8646	847 days

Operating Guidelines and Ranges	
Minimum Suction Pressure (psi) = 632	
- Well on line following workover. Gauge system is Exal, use bender for earth fault detection. - No issues with this well, new pump should start ok and run in current limit. - This pump is shaft limited, it will break above 100 motor amps!!! - Only suction pressure & temp are correct Exal gauge, ignore other data.	
Startup Guidelines and Tips	
- Set frequency to 65hz and start up into current limit, check downhole flow on Exal screen. - Leave well to stabilise and set frequency to final stable freq. + 2hz. - Underload set to 80% of I limit.	

ESP Trip Details at 18 th June 2003			
High Wellhead Trip Pressure (bar) =	90	VSD Underload Current =	756
Low Wellhead Trip Pressure (bar) =	10	VSD Overload Current =	1047

Installed Equipment Details			
Pump:	HN15000 69-COM	Max Motor Amps:	99.4
Trans Ratio:	9.58	Motor:	840HP
(Motor Amps = Drive Amps/Trans Ratio)		Drive Type: C/LIFT 1040	Max Drive Amps: 952

1.5 ESP OPERATIONS

1.5.1 Pump starting

This is the aspect of ESP operation posing the greatest potential risk to the pump, particularly if a site/plant shutdown requires the starting of a number of ESP's in succession (5 or more).

Detailed pump operating limits should be available for reference, local to the motor controller or VSD. Where permanent downhole gauges are employed the data should be also be displayed locally, preferably with a real time trending facility. Downhole data should be relayed to the CCR where as a minimum, alarms should be programmed on; downhole (or motor winding) temperature, earth leakage, and wellhead (high and low) pressure.

Prior to starting the pump, the cause of the trip should be determined and if appropriate rectified. If the cause of the trip cannot be determined one start attempt may be made but repeated starts (on a tripping system) must be avoided. Sufficient time should be allowed between a trip and an attempt to restart for the pressure across the pump to equalise. The fluid column will naturally backflow through a recently stopped pump, causing the unit to backspin. Starting a pump whilst it is backspinning risks overloading the shaft causing torsion failure (at the splined connection between the pump and seal section).

The well should be lined up to the flowline system and the choke partially closed. Floating impellor pumps in particular should be started against a choke to ensure the impellers are seated against their respective downthrust washers; to prevent recycling, increase pump efficiency and avoid impellers running against their upthrust washers, which are small with limited bearing capacity. Once the pump is running, checks should immediately be made to confirm the well is flowing by:

- Comparing running current against the normal motor current.
- Checking for wellhead pressure rise and fall whilst fully opening choke.
- Checking for wellhead temperature rise.

Once the pump is started it may take some time for the wellbore to clean up to steady state conditions. During the wellbore cleanup the pump loading will vary, as soon as this phase is complete the motor controller or VSD should be programmed with an underload based upon the actual loading and frequency of the motor. Setting underload protection prior to establishing steady state flow risks an unnecessary trip and restart on the ESP.

CURRENT LOADING PREDICTIONS IN ESP SYSTEMS

1.6 SECTION INTRODUCTION

This section contains four worked examples of how to calculate the motor current using pump and motor charts and basic well data. The examples are:

- Normal Operation
- Undersized Pump
- Oversized Pump
- Variable Speed Drive

Each example follows a similar format to determine the power required by the pump at a given flow rate using the motor curve to determine current demanded by the motor. The aim is to enable the PE / Well Service Engineer to understand pump electrical performance to help optimise the pump system, troubleshoot problems and design optimum pump systems for future replacement.

The fluid in the pump during operation is at reservoir conditions (approximately) therefore the flowrate must be quoted at reservoir conditions. This document uses reservoir standard cubic metres per day as the flowrate unit, designated as rm^3/d . The gas oil ratio figures are supplied as standard cubic meters of gas per standard cubic metre of oil (scm/scm).

1.7 EXAMPLE 1 - NORMAL OPERATION

The information contained within Section 1.7 is an example of how to calculate the predicted motor current for a given ESP system at a specific flowrate and surface voltage. The example utilises a correctly designed ESP system, i.e. the pump is correctly matched to the well performance and the motor is of suitable size for the pump. This example is therefore intended to illustrate current prediction in a correctly designed ESP system which is operating within its recommended range.

1.7.1 Available Data

Section 1.7.1 contains information which is typically available to the wellsite petroleum engineer, most of which is required for current calculation. The information required and how it is used will become evident as the example progresses.

The following is a summary of the most recent well test data:

Oil Rate (rm ³ /d)	Water Rate (m ³ /d)	GOR (scm/scm)	Reservoir Temperature (degF)	Oil Specific Gravity	ESP Frequency (Hz)
110	0	53	150	0.92	60

Table 1: Example 1 well test data

The table below contains a summary of the ESP system in the well:

Pump Type	Alnas 362 S700
Number of pump stages	242
Motor Type	Borets 32-117 B5
Motor Nameplate Power (kW) @ 60Hz	38.4
Motor Nameplate Voltage (V) @ 60Hz	1080
Motor Nameplate Current (A)	28.9
Power Cable Type	No. 4 Cu
Length of cable from surface switchboard to motor (ft)	7600
Applied Surface Voltage (V)	1172

Table 2: Example 1 ESP data

The pump curve can be seen below in Figure 1, and the composite motor curve can be seen below in Figure 2.

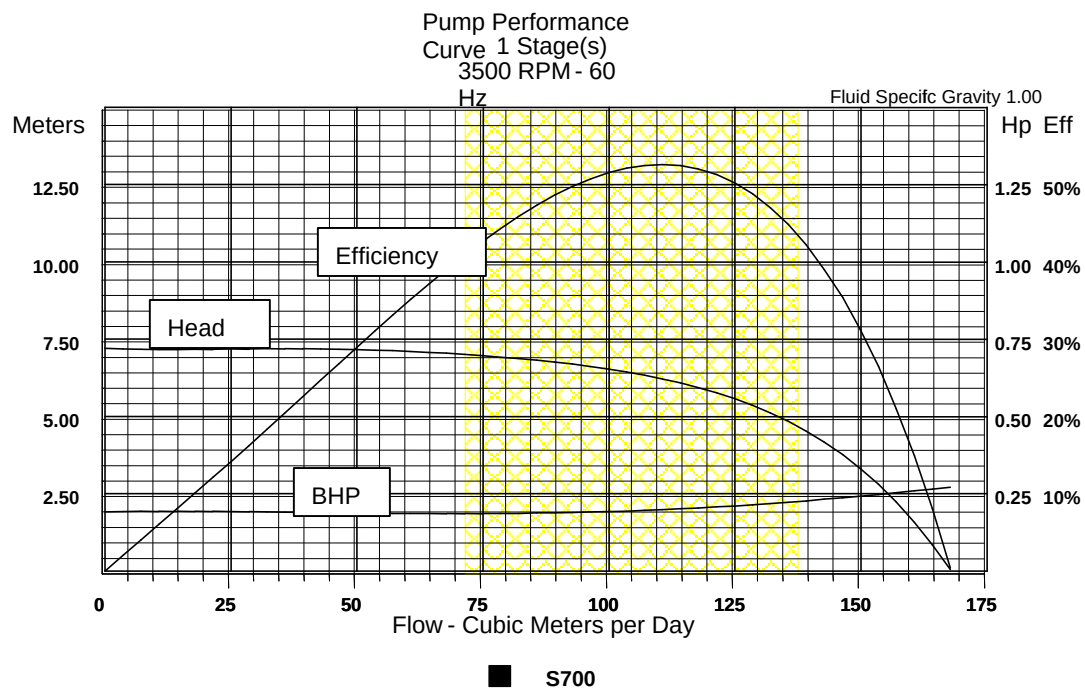


Figure 1: Alnas 362 S700 Pump performance curve for single stage at 60Hz (3500 RPM) and SG=1.0

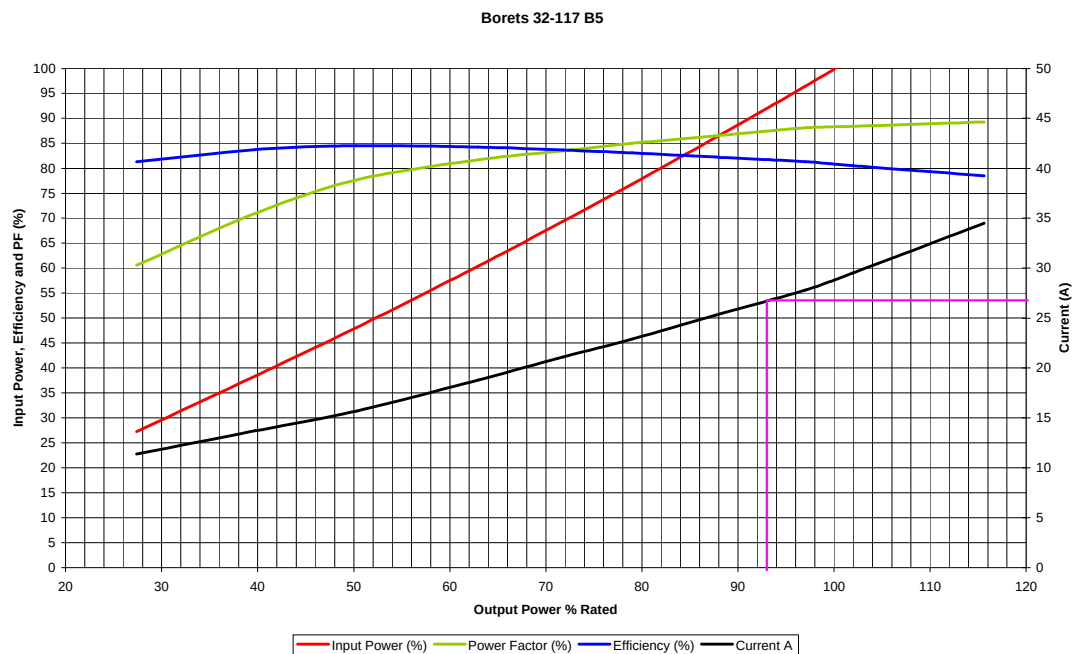


Figure 2: Motor Composite Curve for Borets 32-117 B5 motor

1.7.2 Worked Example Data Flow

The calculation is worked through in detail in the following sections, however a summary of the data flow is presented here for reasons of clarity. The individual steps required to calculate the current in the above ESP system are as follows:

- Determine power required by each pump stage at the specified flowrate. This is achieved using the pump curve
- Multiply power per stage by number of stages to determine total power required by the pump
- Determine power required by protector by using protector loading curve
- Add protector and pump power requirements to obtain total system power requirements. This is equivalent to the motor output power required
- Use motor curve to determine current required by motor to deliver power output.

1.7.3 Determine Power Required by Pump

The first step in determining the required current is to calculate the total power required by the pump to flow the well at the test rate of 110 m³/day. This is calculated using the following formula:

Total pump power required = BHP/stage x fluid SG x No of stages

The pump chart is used to obtain the BHP per stage required for a given flowrate, as is demonstrated by the red solid lines in Figure 3 below. It can be seen that the power required per stage is 0.2 BHP. This is the value of BHP required, at this flowrate, for a single stage at 60 Hz with a fluid SG of 1.0. This is then input into the equation mentioned previously as follows:

$$\begin{aligned}\text{Total pump power required} &= \text{BHP/stage} \times \text{fluid SG} \times \text{No of stages} \\ &= 0.2 \times 0.92 \times 242 \\ &= 44.5 \text{ BHP.}\end{aligned}$$

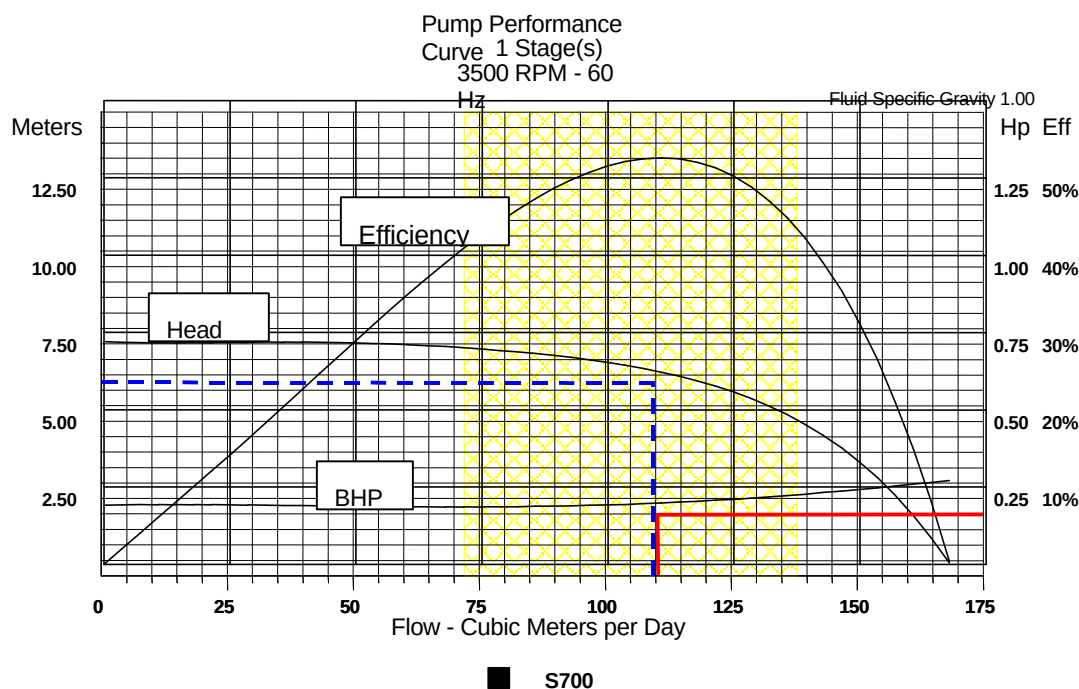


Figure 3: Determining pump stage BHP and head at a specific flowrate

1.7.3.1 Power Required by Protector

The next step in the current calculation is to determine the power which will be consumed by the pump protector, or seal section. To determine the BHP required by the protector, the total head from the pump requires to be established. Referring to the pump curve in Figure 3 above, the head generated per stage (at a specific flowrate at 60 Hz) can be determined as demonstrated by the blue dashed lines. In this example the head per stage is 6.25 metres. As the pump has 242 stages, the total head generated is simply:

Total head = head per stage x No of stages

$$= 6.25 \times 242$$

$$= 1512.5 \text{ metres.}$$

The power consumed by the protector can then be established from the protector loading curve. Referring to the generic protector curve shown below in Figure 4, the power required by the protector in this example is approximately 3.2 BHP, as illustrated by the red lines. Individual component protector curves should be utilised for actual field use, however in the absence of this data for small to medium ESP systems a figure of less than 5 BHP is sufficient.

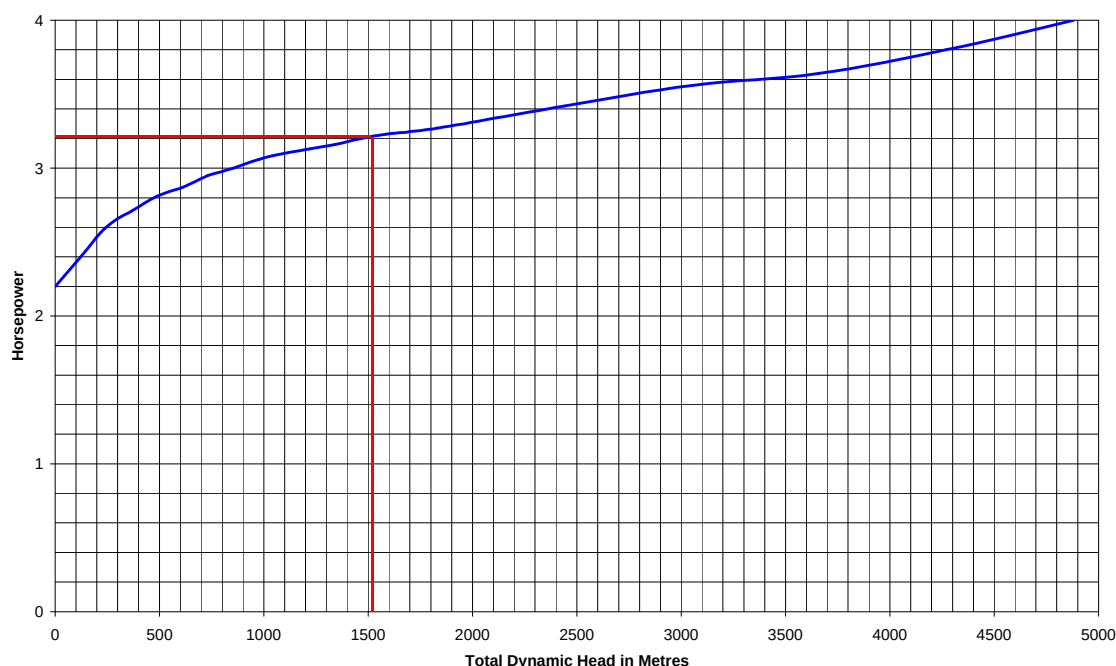


Figure 4: Generic Protector Loading Curve

1.7.4 Total System Power Required

The total power required by the ESP system (i.e. pump and protector) is simply the total of the pump power required and the protector power required. This is as follows:

$$\begin{aligned} \text{Total power required} &= \text{pump power} + \text{protector power} \\ &= 44.5 + 3.2 \\ &= 47.7 \text{ BHP.} \end{aligned}$$

Converting to kW gives:

$$1 \text{ HP} = 0.746 \text{ kW, therefore}$$

$$\begin{aligned} \text{Power in kW} &= 0.746 \times 47.7 \\ &= 35.6 \text{ kW.} \end{aligned}$$

This is the total power input required by the pump system, i.e. the power which is required to be delivered by the motor to the pump.

1.7.5 Determine Motor Current Required

The power required to be delivered by the motor has been calculated above as 35.6 kW. The current required to supply this power can be established in one of two ways:

1. The composite motor curve can be consulted to provide a figure for current. For this method to be effective the voltage supplied to the motor terminals must be the nameplate value at the frequency of operation
2. The current can be calculated from the formula:

$$\text{Current (A)} = [\text{Output power (kW)} \times 1000] / [1.73 \times V \times \text{Eff} \times \text{PF}]$$

Where Eff and PF can be obtained from the motor composite curve.

The simplest manner in which to estimate current required is to use method 1, the motor composite curve. The estimated current is determined by simply reading from the graph the current required for the appropriate motor output power which was previously calculated as 35.6kW. This is equivalent to approx 93% of the motor rating of 38.4kW. The current required can then be simply read from the chart. **The value obtained in this example is approximately 27 Amperes, which is illustrated by the pink lines in Figure 2.**

Note: The calculation of the figure of 27A is the entire focus of this document, i.e. how to use basic well and ESP data to obtain the theoretical value of current which a given ESP system will require for a specified flowrate. It must also be noted that this calculated figure will rarely match the actual measured figure exactly due to the various tolerances and uncertainties involved, e.g. well test and instrument accuracy etc.

1.7.6 Power Cable Losses

To ensure that the nameplate voltage is supplied to the motor terminals, power cable voltage losses, for the relevant cable type, must be calculated and allowed for. This is achieved using a chart as shown below in Figure 5 (for actual field use the cable manufacturer's chart should be used). Using the current figure of 27 A which was obtained previously, a value of voltage drop (per 1000ft of cable) of 10.5 V is obtained (at 77°F), as is illustrated by the black lines in Figure 5.

It is then further necessary to correct for the temperature of the conductor as per Table 3 below. The reservoir temperature in the well is 150°F, so the temperature correction factor is 1.15. The total voltage drop over the cable is given by:

$$\begin{aligned}\text{Voltage drop} &= (\text{Voltage drop per 1000 ft} \times \text{cable length} \times \text{temperature correction})/1000 \\ &= (10.5 \times 7600 \times 1.15)/1000 \\ &= 91.8 \text{ V.}\end{aligned}$$

Therefore the total surface voltage required for this system to ensure nameplate voltage is delivered to the motor is:

$$\begin{aligned}\text{Surface voltage} &= \text{motor nameplate voltage} + \text{cable losses} \\ &= 1080 + 91.8 \\ &\approx 1172 \text{ V.}\end{aligned}$$

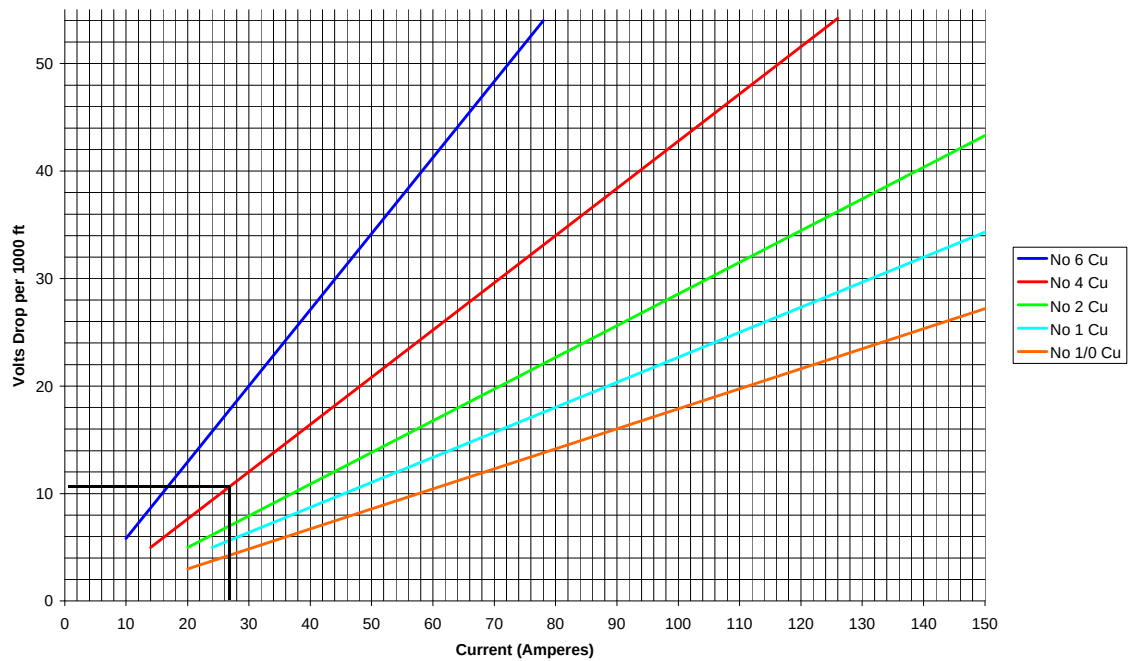


Figure 5: Chart to determine 3 phase Voltage drop in a power cable at 77°F

Conductor Temperature	Voltage Drop Multiplier
131°F (55°C)	1.12
149°F (65°C)	1.15
167°F (75°C)	1.19
185°F (85°C)	1.23
203°F (95°C)	1.27
221°F (105°C)	1.31
239°F (115°C)	1.35
257°F (125°C)	1.39
275°F (135°C)	1.42
293°F (145°C)	1.46
302°F (150°C)	1.48

Table 3: Voltage drop multipliers

1.7.7 Conclusions

- This is a well designed system which is operating close the pump best efficiency point (BEP)
- The motor is well matched to the pump in that it is operating relatively close to (and less than) the motor nameplate power
- The ESP system is appropriately designed and sized for the well inflow performance.

1.8 EXAMPLE 2 - UNDERSIZED PUMP

The following is an example of current calculation in an undersized pump system, i.e. the pump has insufficient flow capacity for the well inflow performance. This example uses the same equipment as example 1 but in a different well.

1.8.1 Available Data

The following is a summary of the most recent well test data:

Oil Rate (rm ³ /d)	Water Rate (m ³ /d)	GOR (scm/scm)	Reservoir Temperature (degF)	Oil Specific Gravity	ESP Frequency (Hz)
155	0	90	185	0.92	60

Table 4: Example 2 well test data

The table below contains a summary of the ESP system in the well:

Pump Type	Alnas 362 S700
Number of pump stages	242
Motor Type	Borets 32-117 B5
Motor Nameplate Power (kW) @ 60Hz	38.4
Motor Nameplate Voltage (V) @ 60Hz	1080
Motor Nameplate Current (A)	28.9
Power Cable Type	No. 4 Cu
Length of cable from surface switchboard to motor (ft)	9200
Applied Surface Voltage (V)	1238

Table 5: Example 2 ESP data

The pump curve can be seen below in Figure 6, and the composite motor curve can be seen below in Figure 7.

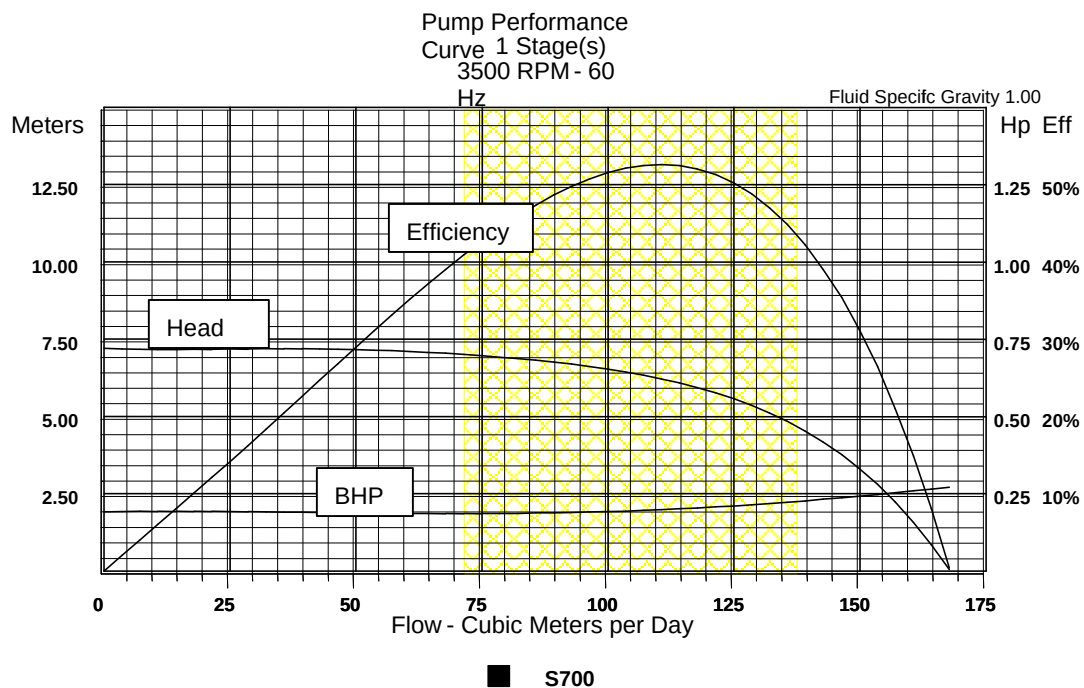


Figure 6: Alnas 362 S700 Pump performance curve for single stage at 60Hz (3500 RPM) and SG=1.0

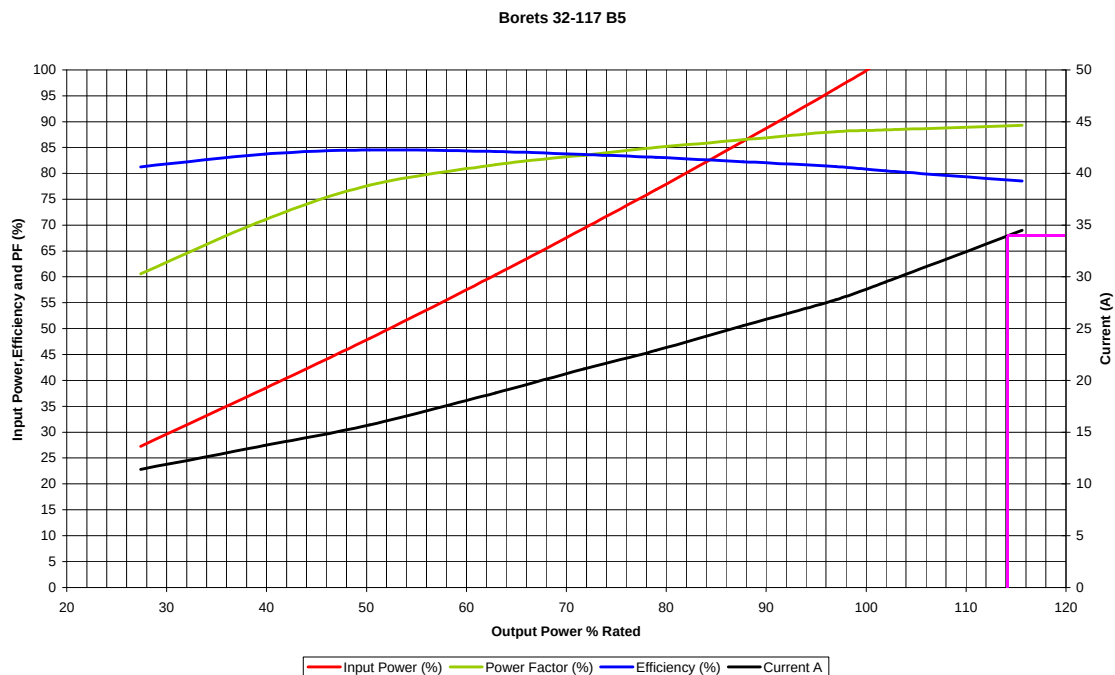


Figure 7: Motor Composite Curve for Borets 32-117 B5 motor

1.8.2 Determine Power Required by Pump

As with example 1, the first step in determining the required current is to calculate the total power required by the pump to flow the well at the test rate of 155 m³/day.

In an identical manner to example 1, the pump chart is used to obtain the BHP required and head generated per stage. As illustrated by the red solid lines in Figure 8 below the BHP per stage in this example is 0.25 BHP. The total pump power required is then:

$$\begin{aligned}\text{Total pump power required} &= \text{BHP/stage} \times \text{fluid SG} \times \text{No of stages} \\ &= 0.25 \times 0.92 \times 242 \\ &= 55.7 \text{ BHP.}\end{aligned}$$

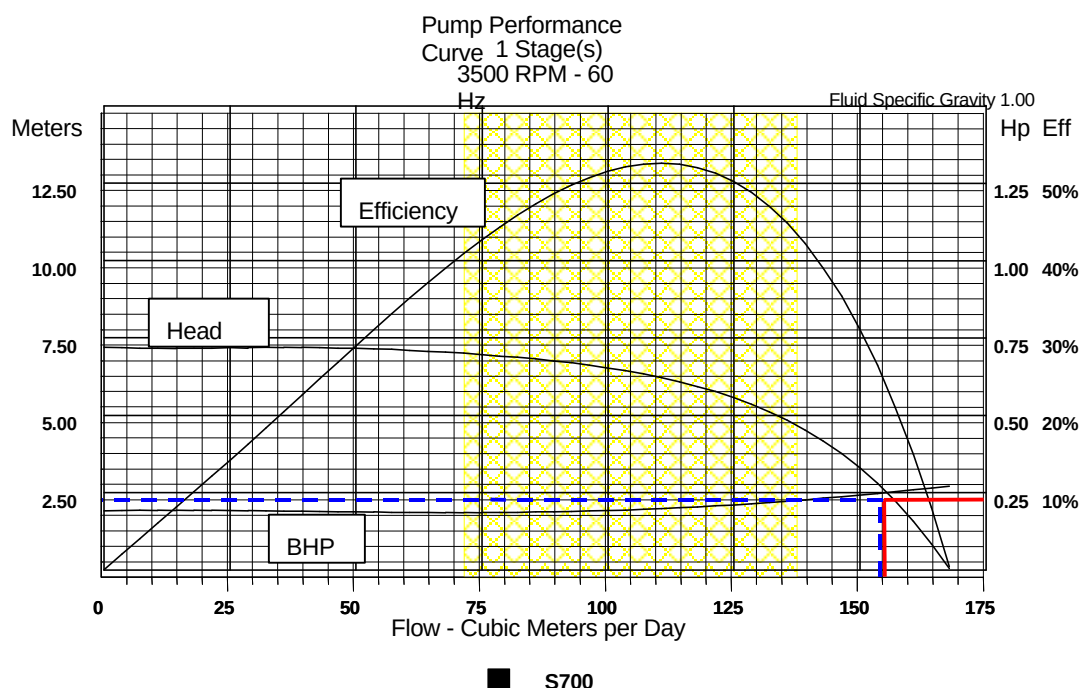


Figure 8: Determining pump stage BHP and head at a specific flowrate

1.8.2.1 Power Required by Protector and Total System Power Required

The head generated per stage in this example is determined as approximately 2.5 metres, as illustrated by the blue dashed lines on Figure 8. The total head generated is therefore:

$$\begin{aligned}\text{Total head} &= \text{head per stage} \times \text{No of stages} \\ &= 2.5 \times 242 \\ &= 605 \text{ metres.}\end{aligned}$$

Referring to the generic protector loading curve shown in Figure 4, the power required by the protector is determined as approximately 2.9 HP.

The total power required by the pump and protector is therefore

$$\begin{aligned}\text{Total power} &= \text{pump power} + \text{protector power} \\ &= 55.7 + 2.9 \\ &= 58.6 \text{ HP.}\end{aligned}$$

Converting this to kW gives $58.6 \times 0.746 = 43.7 \text{ kW}$.

1.8.3 Determine Motor Current Required

The required current is determined in exactly the same manner as for example 1. The motor power requirement has been calculated as 43.7kW, which is equivalent to 114% of the motor rating of 38.4kW. As illustrated in Figure 7 by the pink lines, it can be seen that a value of approximately **34 Amperes** is obtained for this power requirement.

The current required by this ESP system operating in this configuration is 34A.

1.8.4 Power Cable Losses

As per example 1, the chart shown in Figure 5 is consulted to obtain the voltage drop per 1000 ft of cable, and in this instance for a current of 34 A, a figure of approximately 14 V per 1000 ft is obtained for a No 4 cable.

The reservoir temperature is 185°F, therefore from Table 3 the temperature correction is 1.23.

This data is then used to calculate the cable voltage drop as below:

$$\text{Voltage drop} = (\text{Voltage drop per 1000 ft} \times \text{cable length} \times \text{temperature correction})/1000$$

$$\begin{aligned}\text{Voltage drop} &= (14 \times 9200 \times 1.23)/1000 \\ &= 158.4 \approx 158 \text{ V.}\end{aligned}$$

The voltage applied to the motor terminals is then

$$\begin{aligned}\text{Motor voltage applied} &= \text{surface voltage} - \text{cable voltage drop} \\ &= 1238 - 158 \\ &= 1080 \text{ V.}\end{aligned}$$

therefore the correct voltage is being applied to the motor terminals.

1.8.5 Conclusions

The following are the key observations regarding this ESP system. For a detailed explanation refer to section 2.2.

- The pump is operating in an upthrust condition
- The motor is overloaded – it is running at approximately 118% nameplate current
- Damage will occur to either the pump, motor or both, resulting in reduced run lifetime
- The pump in this well is of insufficient capacity. If possible it should be replaced with a unit with a greater flow capacity, together with a motor suitable for the power requirements
- If the current overload has been set at 110% nameplate, the system will not run – it will trip as soon as flowing conditions have been established.

1.9 EXAMPLE 3 - OVERSIZED PUMP

The final fixed speed example is a current calculation in an oversized system, i.e. the pump capacity is greater than the well inflow performance can deliver.

This example uses the same well as example 1, which had a correctly designed system which was matched to the well inflow performance, resulting in efficient ESP operation. In this example it will be assumed that the equipment was pulled from the well (for whatever reason), and replaced with inappropriate and poorly matched equipment, to demonstrate the contrasting power and current requirements.

1.9.1 Available Data

The following is a summary of the most recent well test data:

Oil Rate (rm ³ /d)	Water Rate (m ³ /d)	GOR (scm/scm)	Reservoir Temperature (degF)	Oil Specific Gravity	ESP Frequency (Hz)
110	0	53	150	0.92	60

Table 6: Example 3 well test data

The table below contains a summary of the ESP system in the well:

Pump Type	Alnas 362 M3000
Number of pump stages	166
Motor Type	Borets 90-117 B5
Motor Nameplate Power (kW) @ 60Hz	108
Motor Nameplate Voltage (V) @ 60Hz	2400
Motor Nameplate Current (A)	36.3
Power Cable Type	No. 4 Cu
Length of cable from surface switchboard to motor (ft)	7600
Applied Surface Voltage (V)	2505

Table 7: Example 3 ESP data

The pump curve can be seen below in Figure 9, and the composite motor curve can be seen below in Figure 10.

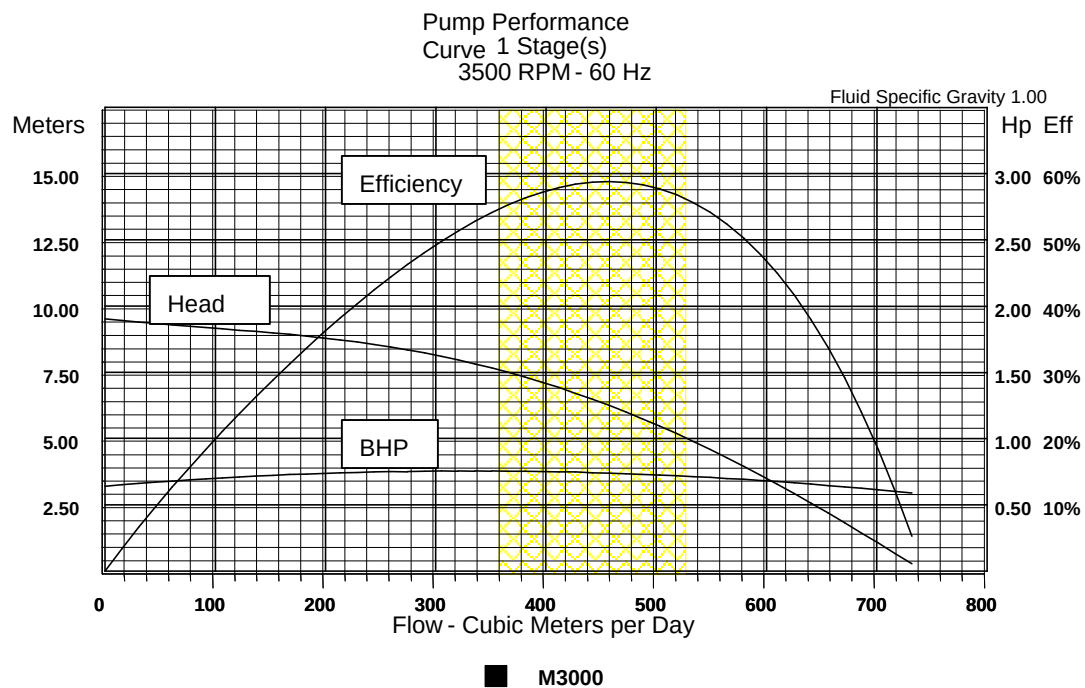


Figure 9: Alnas 362 M3000 Pump performance curve for single stage at 60Hz (3500 RPM) and SG=1.0

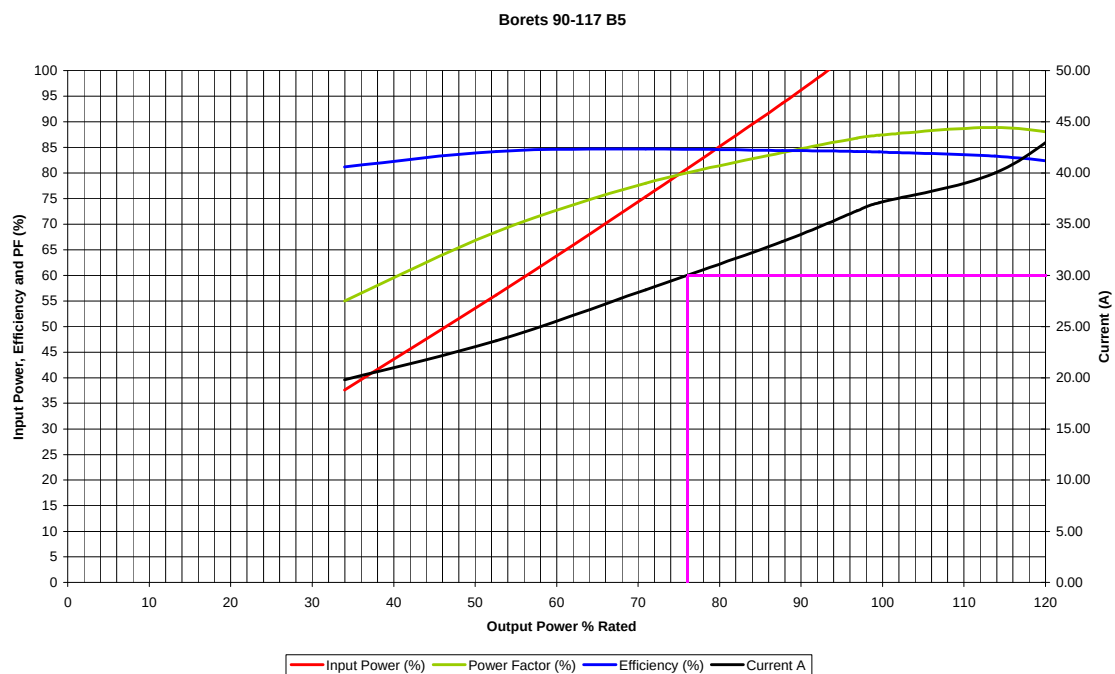


Figure 10: Motor Composite Curve for Borets 90-117 B5 motor

1.9.2 Determine Power Required by Pump

As with the previous examples, the pump chart is used to obtain the BHP required and head generated per stage. As illustrated by the red solid lines in Figure 11 below the BHP per stage in this example is 0.7 BHP. The total pump power required is then:

$$\begin{aligned}\text{Total pump power required} &= \text{BHP/stage} \times \text{fluid SG} \times \text{No of stages} \\ &= 0.7 \times 0.92 \times 166 \\ &= 107 \text{ BHP.}\end{aligned}$$

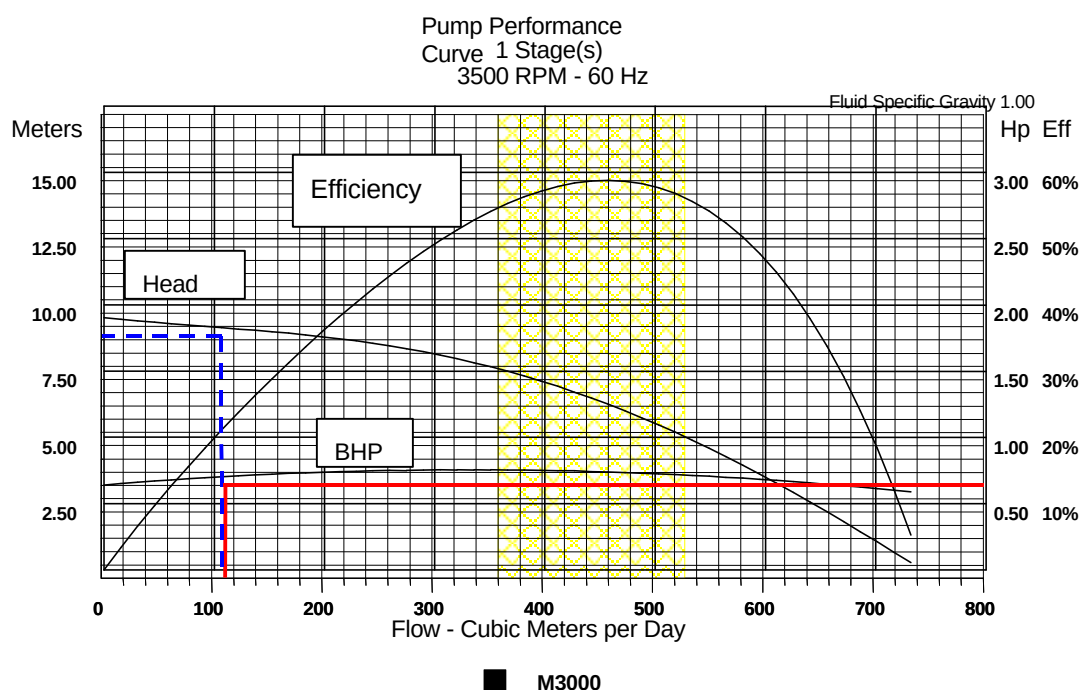


Figure 11: Determining pump stage BHP and head at a specific flowrate

1.9.2.1 Power Required by Protector and Total System Power Required

Referring to Figure 11 above, the head generated per stage is 9.1 metres, as illustrated by the blue dashed lines. As the pump has 166 stages, the total head generated is:

$$\begin{aligned}\text{Total head} &= \text{head per stage} \times \text{No of stages} \\ &= 9.1 \times 166 \\ &\approx 1511 \text{ metres.}\end{aligned}$$

Referring to the generic protector loading curve shown in Figure 4, the power required by the protector is approximately 3.2 HP.

The total power required by the pump and protector is therefore

$$\begin{aligned}\text{Total power} &= \text{pump power} + \text{protector power} \\ &= 107 + 3.2 \\ &= 110.2 \text{ HP.}\end{aligned}$$

Converting this to kW gives $110.2 \times 0.746 \approx 82.2 \text{ kW}$.

1.9.3 Determine Motor Current Required

The required current is determined in the same manner as the previous examples. The motor power requirement has been calculated as 82.2kW, which is equivalent to 76% of the motor rating of 108kW. As illustrated in Figure 10 by the pink lines, a value of approximately **30 Amperes** is obtained for the power requirement of 82.2kW previously calculated.

The current required by this ESP system operating in this configuration is 30A.

1.9.4 Power Cable Losses

The chart shown in Figure 5 is consulted to obtain the voltage drop per 1000 ft of cable, and in this example for a current of 30 A, a figure of approximately 12 V per 1000 ft is obtained for a No 4 cable.

The reservoir temperature is 150°F, therefore from Table 3 the temperature correction is 1.15.

This data is then used to calculate the cable voltage drop as below:

$$\text{Voltage drop} = (\text{Voltage drop per 1000 ft} \times \text{cable length} \times \text{temperature correction}) / 1000$$

$$\begin{aligned}\text{Voltage drop} &= (12 \times 7600 \times 1.15) / 1000 \\ &= 104.9 \approx 105 \text{ V.}\end{aligned}$$

The voltage applied to the motor terminals is then

$$\begin{aligned}\text{Motor voltage applied} &= \text{surface voltage} - \text{cable voltage drop} \\ &= 2505 - 105 \\ &= 2400 \text{ V.}\end{aligned}$$

therefore the correct voltage is being applied to the motor terminals.

1.9.5 Conclusions

The following are the key observations regarding this ESP system. For a detailed explanation refer to section 2.3.

- The pump is operating in a downthrust condition and will most likely experience a shortened run life
- The pump in this well has a flow capacity greater than that required by the well inflow performance. If possible the pump should be replaced with a unit with a reduced flow capacity, together with a suitable sized motor
- The system is so badly designed that a totally different motor of almost 3 times the power rating has to be used compared to the same well with a correctly designed system (example 1).

1.10 EXAMPLE 4 - VARIABLE SPEED SYSTEM

This example uses an ESP system which has a variable speed controller which can operate from 35 Hz to 60 Hz, and is intended to demonstrate current calculation in a variable speed system which is operating at some frequency other than the maximum design. It is also intended to demonstrate how to convert the data contained within pump and motor charts from the base frequency to the actual frequency of operation. The effects and potential benefits of variable speed systems will then be further discussed in Section 2.4.

1.10.1 Available Data

The following is a summary of the most recent well test data:

Oil Rate (rm ³ /d)	Water Rate (m ³ /d)	GOR (scm/scm)	Reservoir Temperature (degF)	Oil Specific Gravity	ESP Frequency (Hz)
600	0	40	130	0.8	53

Table 8: Example 3 well test data

The table below contains a summary of the ESP system in the well:

Pump Type	Alnas 362 M3800
Number of pump stages	155
Motor Type	Borets 63-117 B5
Motor Nameplate Power (kW) @ 60Hz	75.6
Motor Nameplate Voltage (V) @ 60Hz	2400
Motor Nameplate Current (A)	26.2
Power Cable Type	No. 2 Cu
Length of cable from surface VSD to motor (ft)	4000
Applied Surface Voltage (V)	2148

Table 9: Example 3 ESP data

The pump curve can be seen below in Figure 12, and the composite motor curve can be seen below in Figure 13.

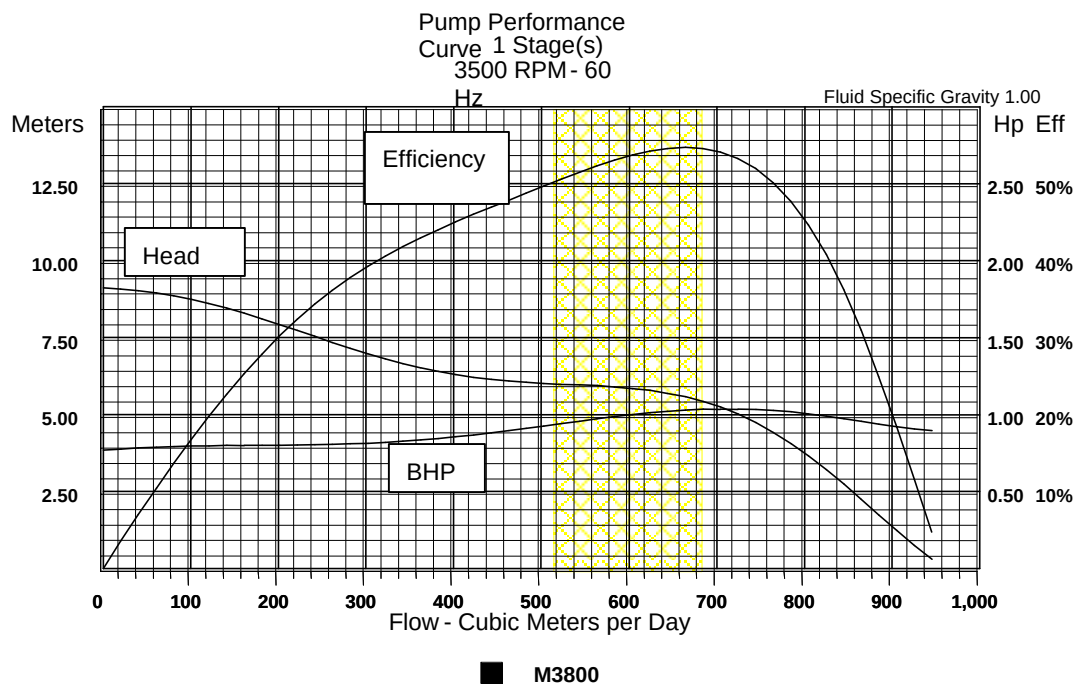


Figure 12: Alnas 362 M3800 Pump performance curve for single stage at 60Hz (3500 RPM) and SG=1.0

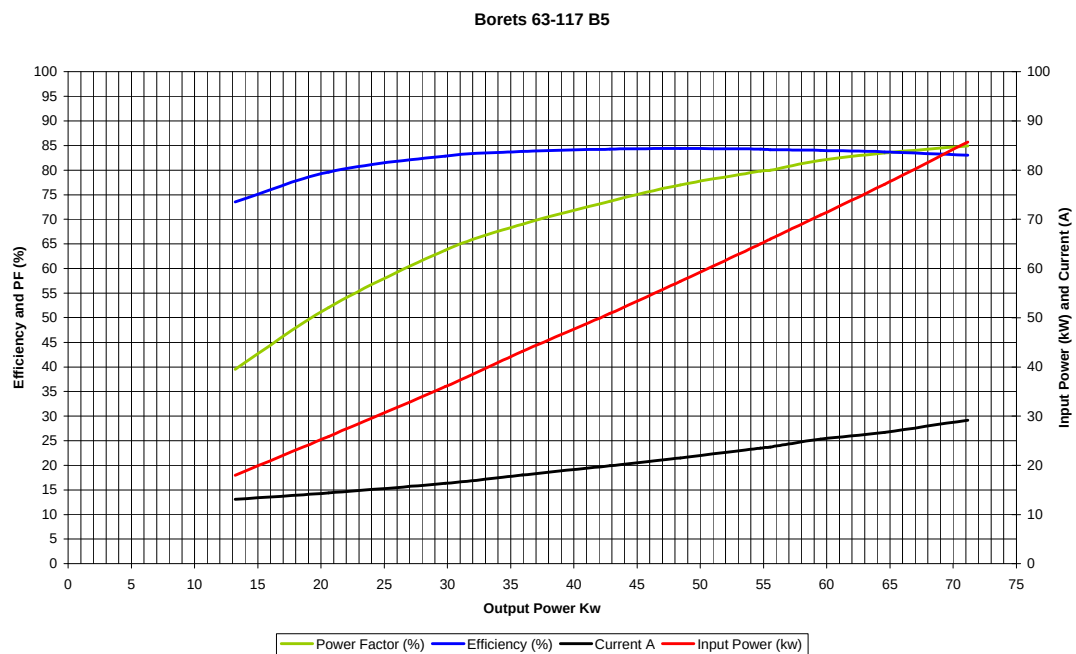


Figure 13: Motor Composite Curve for Borets 63-117 B5 motor at 60Hz

1.10.2 Determine Power Required by Pump

The pump curve in Figure 12 is for an operating frequency of 60Hz. In this instance the operating frequency of the ESP system is 53Hz, therefore it is apparent that this curve is not applicable for this situation. The BHP per stage is calculated in the following manner:

1. Convert the 53Hz flowrate to a 60Hz equivalent flowrate
2. Determine the 60Hz equivalent BHP required per pump stage
3. Convert the 60Hz figure to an equivalent 53Hz value.

This will be outlined in the following sections.

1.10.2.1 Convert 53Hz Flowrate to 60Hz Equivalent

It is necessary to convert the 53Hz flowrate to a 60Hz equivalent to ensure that the BHP per stage figure is read from the correct part of the BHP curve, as the BHP per stage varies at differing flowrates. This is performed by applying one of a series of equations known as the affinity laws to the 53Hz flowrate as follows:

Flowrate 1 = (Frequency 1/Frequency 2) x Flowrate 2, i.e.

$$\begin{aligned}\text{Flowrate}_{60\text{Hz}} &= (60/53) \times 600 \\ &\approx 680 \text{ m}^3/\text{d}.\end{aligned}$$

Care must be taken when applying this equation; this is discussed further in Section 2.4.

1.10.2.2 Determine BHP Required per Stage for 60Hz Operation

The BHP required per stage is then determined assuming 60Hz pump operation (as this is the frequency at which the chart available is based upon). The equivalent 60Hz flowrate has been calculated as 680 m³/d, therefore the BHP per stage is determined, as per previous examples, using the pump chart (a 60Hz chart is used assuming that this is the only chart available to the engineer) as illustrated by the red solid lines in Figure 14 below. It can be seen that for a flowrate of 680m³/d at 60Hz pump operation the BHP per stage required would be approx 1.03 BHP.

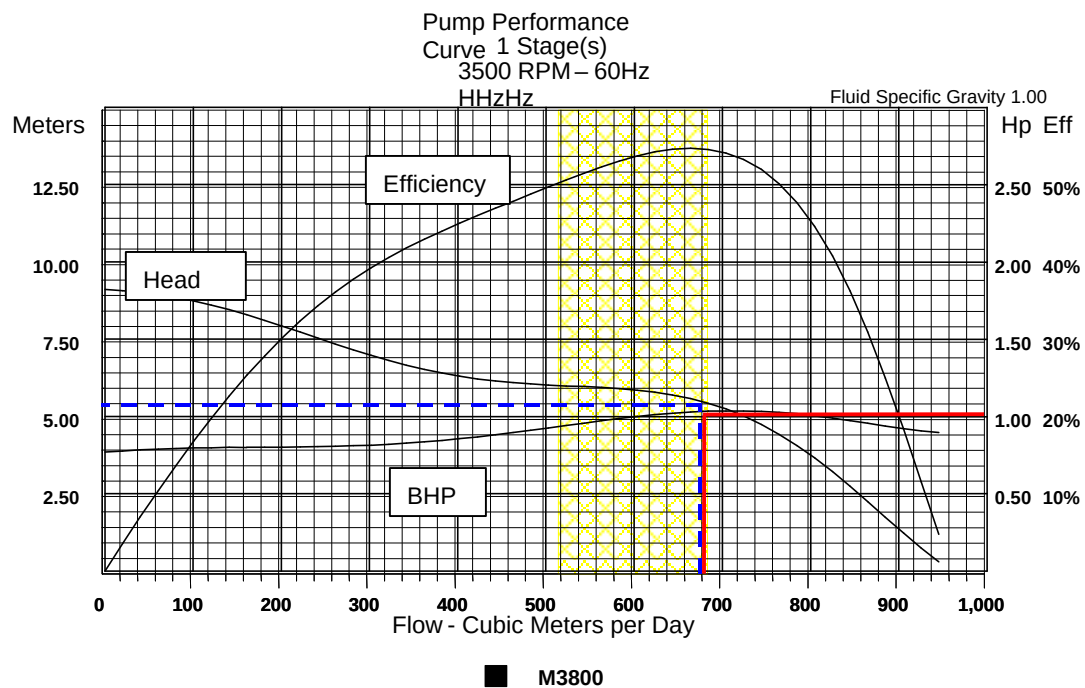


Figure 14: Determining pump stage BHP and head at a specific flowrate for 60Hz operation

1.10.2.3 Convert 60Hz BHP Figure to an Equivalent 53Hz Value

The BHP per stage assuming 60Hz ESP operation has been calculated as 1.03 BHP. To convert this to an equivalent figure for actual system operation of 53Hz, i.e. to calculate the actual pump requirement, the second of the affinity laws is applied to the 60Hz value:

$$\text{BHP 1} = (\text{Frequency 1/Frequency 2})^3 \times \text{BHP 2}$$

where BHP 1 is at frequency 1 and BHP 2 is at frequency 2.

Therefore the actual pump BHP requirement per stage at 53 Hz is calculated as follows:

$$\begin{aligned}\text{BHP}_{53\text{Hz}} &= (53/60)^3 \times 1.03 \\ &\approx 0.69 \text{ BHP/stage.}\end{aligned}$$

The total pump power required is then:

$$\begin{aligned}\text{Total pump power required} &= \text{BHP/stage} \times \text{fluid SG} \times \text{No of stages} \\ &= 0.69 \times 0.8 \times 155 \\ &\sim 85.6 \text{ BHP.}\end{aligned}$$

1.10.3 Power Required by Protector and Total System Power Required

Referring again to Figure 14, it can be seen that the head generated per stage, for an equivalent flowrate of 680m³/d and at a frequency of 60Hz, is approx 5.5 metres, as illustrated by the blue dashed lines. This head per stage value is also required to be converted to an equivalent 53Hz figure. This is achieved using the third and final affinity law:

$$\text{Head 1} = (\text{Frequency 1/Frequency 2})^2 \times \text{Head 2}$$

Therefore the actual head per stage at 53Hz is:

$$\begin{aligned}\text{Head}_{53\text{Hz}} &= (53/60)^2 \times 5.5 \\ &\sim 4.3 \text{ m.}\end{aligned}$$

As the pump has 129 stages, the total head generated is:

$$\begin{aligned}\text{Total head} &= \text{head per stage} \times \text{No of stages} \\ &= 4.3 \times 155 \\ &\sim 666.5 \text{ metres.}\end{aligned}$$

Referring to the generic protector loading curve shown in Figure 4, the power required by the protector is approximately 2.9 HP.

The total power required by the pump and protector is therefore

$$\begin{aligned}\text{Total power} &= \text{pump power} + \text{protector power} \\ &= 85.6 + 2.9 \\ &= 88.5 \text{ HP.}\end{aligned}$$

Converting this to kW gives $88.5 \times 0.746 \approx 66 \text{ kW}$.

1.10.4 Determine Motor Current Required

The current drawn by the motor is determined in a similar manner as the previous examples, i.e. from reading off the motor chart. However there are some important differences between operation of a fixed speed system and a variable speed system which must be briefly outlined. The first point is that operation at different frequencies requires that the voltage supplied to the motor is varied directly with frequency. The second point is that a motor does not have the same power output at all frequencies – again this varies directly with frequency. The motor in this example is rated at 75.6kW at 60Hz. Operation of this motor at 53Hz results in a new power output rating of the motor as follows:

$$\text{Power Output 1} = (\text{Frequency 1/Frequency 2}) \times \text{Power Output 2}$$

i.e. the power output of a motor varies directly as frequency.

Therefore in this example, the power rating of the motor at 53 Hz is:

$$\begin{aligned}\text{Power Output}_{53\text{Hz}} &= (53/60) \times 75.6 \\ &= 66.8 \text{ kW} = 89.5 \text{ HP}.\end{aligned}$$

Another important point is that when a motor is operated at a different frequency with the voltage varied directly as frequency the current rating of the motor does not change with frequency. Therefore in this example, it has been established that the power rating of the motor is 89.5 HP at 53Hz, so it follows that if the motor load is 89.5 HP at 53Hz then the current required by the motor is the same value as the nameplate current quoted at 60Hz, i.e. 26.2 A.

To obtain the current required, it is necessary to use a motor chart which is scaled in terms of percentage of rated values (for fixed speed systems the chart can be scaled in absolute values or percentages). Thus, in this example, the 100% output power value at 53Hz would correspond to the figure of 89.5 HP, quoted above. The current magnitude can then be simply read from the chart, as in the previous examples, as the current rating does not change with frequency. In this case the power required by the pump, and thus power output required from the motor, is 88.5 HP. The motor rating at this frequency has been calculated at 89.5 HP, therefore the motor output is $(88.5/89.5) \times 100 = 99\%$ of the rated value at 53Hz. The current required can then be simply read from the rescaled motor chart, which is illustrated by the pink lines in Figure 15 below.

It can be seen that a value of approximately **26 Amperes** is obtained for the power requirement of 66 kW (or 99% rated load at 53Hz) previously calculated. This is as we would expect given that the motor output is 99% of that rated, therefore the current will be approximately nameplate value.

The current required by this ESP system operating in this configuration is 26 A.

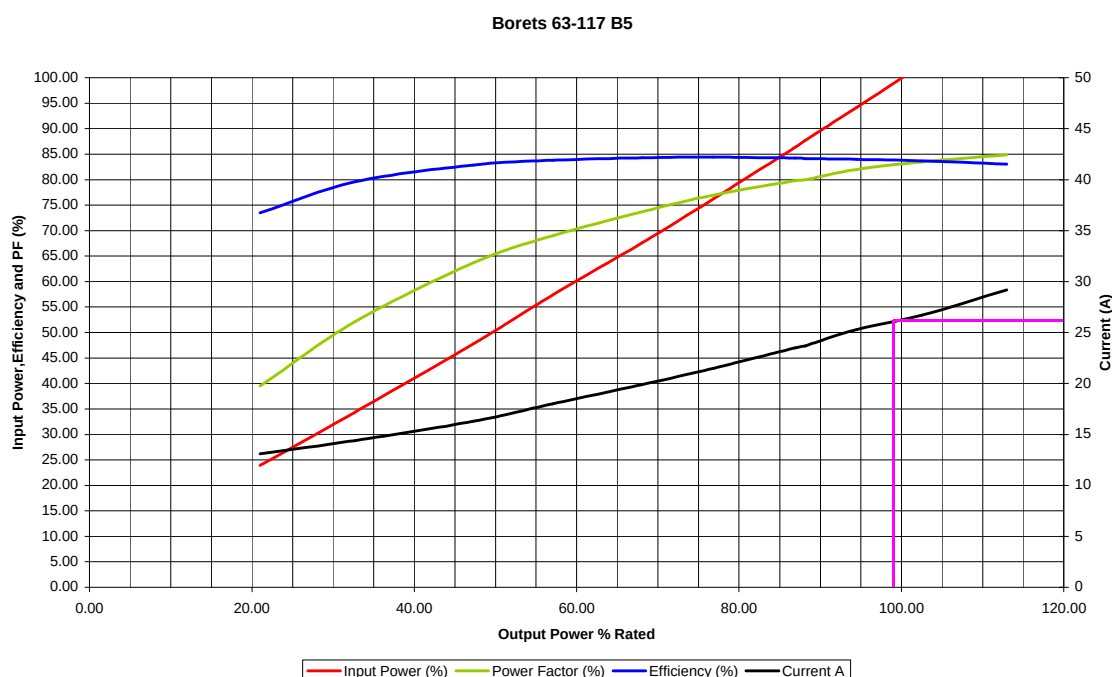


Figure 15: Current estimation from % rated motor chart

It should be noted here that this technique is not strictly accurate as motor efficiency increases with frequency, i.e. the efficiency at 53Hz will be less than that at 60Hz, however this effect is negligible at the typical frequencies encountered in ESP systems.

1.10.5 Power Cable Losses

The chart shown in Figure 5 is consulted to obtain the voltage drop per 1000 ft of cable, and in this example for a current of 26 A, a figure of approximately 6.3 V per 1000 ft is obtained for a No 2 cable.

The reservoir temperature is 130°F, therefore from Table 3 the temperature correction is 1.12.

This data is then used to calculate the cable voltage drop as below:

$$\begin{aligned}\text{Voltage drop} &= (\text{Voltage drop per 1000 ft} \times \text{cable length} \times \text{temperature correction})/1000 \\ \text{Voltage drop} &= (6.3 \times 4000 \times 1.12)/1000 \\ &= 28.2 \approx 28 \text{ V.}\end{aligned}$$

It was previously mentioned that in varying the frequency of an ESP system the voltage is varied directly as the frequency. Therefore the voltage actually supplied to a 60Hz motor operating at 53Hz is $(53/60) \times \text{nameplate voltage}$

$$\begin{aligned}&= (53/60) \times 2400 \\ &= 2120 \text{ V.}\end{aligned}$$

The voltage applied to the motor terminals is then

$$\begin{aligned}\text{Motor voltage applied} &= \text{surface voltage} - \text{cable voltage drop} \\ &= 2148 - 28 \\ &= 2120 \text{ V.}\end{aligned}$$

therefore the correct voltage is being applied to the motor terminals.

1.10.6 Conclusions

The following are the key observations regarding this ESP system. For a detailed analysis of VSD (variable speed drive) systems refer to section 2.4.

- In a variable speed system the principles of current calculation are similar in principle to fixed speed systems, however a few important differences exist
- Power required and head generated per pump stage are determined using the available pump chart then the values obtained are converted, using affinity law equations, to equivalent values for the actual frequency of ESP operation
- When a variable speed system is used, the applied motor voltage is varied directly as frequency
- The power rating of a motor varies directly with the applied frequency
- The current required by the motor is read from a rescaled chart in terms of percentage rated motor power output
- The pump in this system is operating close the best efficiency point (BEP)
- The motor in this system is operating at around 99% nameplate power.

2.0 TROUBLESHOOTING

This section examines how to apply the information from the worked examples in Section 3.0 to operational problems and is structured using the same sub-categories for convenience.

The focus is on operational problems impacting the electrical system including motor overload; transformer settings; well pump-off and gas locking; underload protection; dead heading and effect of VSD frequency changes on ESP performance. A brief section on benefits of VSD systems is also included.

2.1 NORMAL OPERATION - GENERAL

The current required for the ESP system in example 1, at flowing conditions as per Table 1, is approximately 27 A. This is approximately 93% of the motor nameplate current of 28.9 A. It must be appreciated that the current predicted will not be a precise value, considering the small tolerances which are expected given the graphical methodology of determining the current (and also the inherent data uncertainties involved in oilfield data gathering, e.g. flowrate measurement tolerance etc).

This is an example of a pump which is correctly matched to the well productivity, and also of a motor which is suitably sized for the pump, i.e. both pump and motor are operating at high efficiencies; the pump is close to the BEP and the motor is operating close to nameplate power. In reality this sort of close match may not be achieved very often, however this serves to illustrate the type of operation that is possible in an ESP system. The case may be that it is desirable to operate at a slightly lower motor power requirement for longevity reasons, this is determined on a well by well basis.

2.1.1 Power Cable Losses

As previously stated, the method of predicting current from the motor composite curve requires that voltage delivered to the downhole motor terminals is nameplate value; 1080 V in the case of example 1. To achieve this voltage at the motor terminals, voltage losses in the power cable from surface to the motor must be calculated and allowed for.

Voltage losses are a result of the natural resistance of the electric cable to current flow. This is analogous to frictional pressure losses in a fluid carrying pipe, i.e. voltage can be considered equivalent to pressure, and current can be considered equivalent to fluid flow. The particular cable used in an ESP application will depend on several factors, including the expected amperage, voltage drop and space available between the tubing and casing. In general, the cable should be selected to give less than 30 volts drop per 1000 ft of cable.

2.2 UNDERSIZED PUMP

Example 2 illustrates the following important concepts in ESP operations:

- Upthrust
- Overloaded Motor
- Corrective actions which may be taken when the two conditions above exist
- ESP design considerations when IPR uncertainties are present.

These issues are further discussed in the sections below.

2.2.1 Pump Operating Conditions – Upthrust

Key Point:

- Pump operation outside the recommended range will most likely result in damage to the pump and subsequent premature failure.

The first observation in example 2 is that the pump is not operating within its recommended envelope. In the pump curve of Figure 8, the recommended operating range is the shaded area in the centre of the figure, ranging approximately from 70 to 140 m³/d on the x-axis. If a pump is operating to the right of this interval, as in this instance, the pump is said to be in upthrust, whereas if the pump is operating to the left of this interval the pump is said to be in downthrust.

The overall thrust on an impeller is the net resulting force of the upward acting fluid pressure on the base of the impeller, the downward acting fluid pressure on the top of the impeller and the upward acting momentum of the fluid within the stage. When operating in the recommended range, most stages are designed to be in light to moderate downthrust. Downthrust increases as the flow through the stage decreases (or as the operating point moves toward the left-hand side of the pump curve). The method of handling pump thrust varies depending on whether fixed or floating impellers are employed, however the net result is that operation in severe downthrust or upthrust will have a negative impact on pump longevity.

This pump is operating in upthrust conditions, which, if not addressed, will most likely result in premature failure of the thrust bearings or some other component of the pump.

2.2.2 Overloaded Motor

Key Points:

- Overload protection is designed to protect the motor from overloading and overheating and is generally set at 110% nameplate current value
- If a system repeatedly trips on overload the setting should not be simply raised, the root cause of the problem should be addressed.

The second problem with this ESP system is that the motor is overloaded. The nameplate current is the maximum recommended current at which the motor should operate, therefore in a correctly designed ESP application the motor will operate at some value of current less than nameplate. In this instance the motor is operating at approximately 34 A, which is equivalent to 118% of nameplate. This is a result of the power capabilities of the motor and the hydraulic power requirements of the pump being considered separately as opposed to being part of one

system, and consequently the two requirements have not been considered in tandem and are poorly matched. This situation presents two main problems:

- The excessive current will most likely result in the premature failure of the motor
- The current is above the level which should cause overload protection trips

The switchboard, or surface power controller, will have protection trips set at pre-established levels. Overload trips are generally set at 110% nameplate current, therefore the current levels experienced in the system described above are greater than this level. If the overload in this system has been set at 110% nameplate current it is apparent that it will immediately trip once flow has been established. Simply raising the overload until the system does not trip will result in a system which is operating outside its design limits and will result in premature motor failure. Potential action to alleviate this problem is described in the next section.

2.2.3 Corrective Action Possible

Key Points:

- Upthrust conditions can be alleviated by choking the production rate back
- Choking the well will most likely also result in the BHP requirements decreasing, therefore reducing the motor loading
- Restricting flowrate is uneconomic; a correctly designed system would allow the well to deliver maximum production.

As discussed above, the ESP system in this example has been poorly designed and is operating in a manner which is likely to result in premature failure of the pump or motor or both. The manner in which this should be addressed is relatively simple. As the pump is operating in upthrust, or on the right hand side of the recommended operating range on the pump curve, it can be appreciated that a shift to the left on this curve would result in the pump at least operating in an improved condition. The pump curve x-axis parameter is flowrate, therefore it can be seen that to move the operating point of the pump to the left, a reduction in flowrate is necessary. This can be achieved by simply choking back the well, resulting in an increased wellhead pressure and a reduced flowrate. Shifting the operating point by choking the well back to flow at 110 m³/d is illustrated in the pump curve in Figure 16 below:

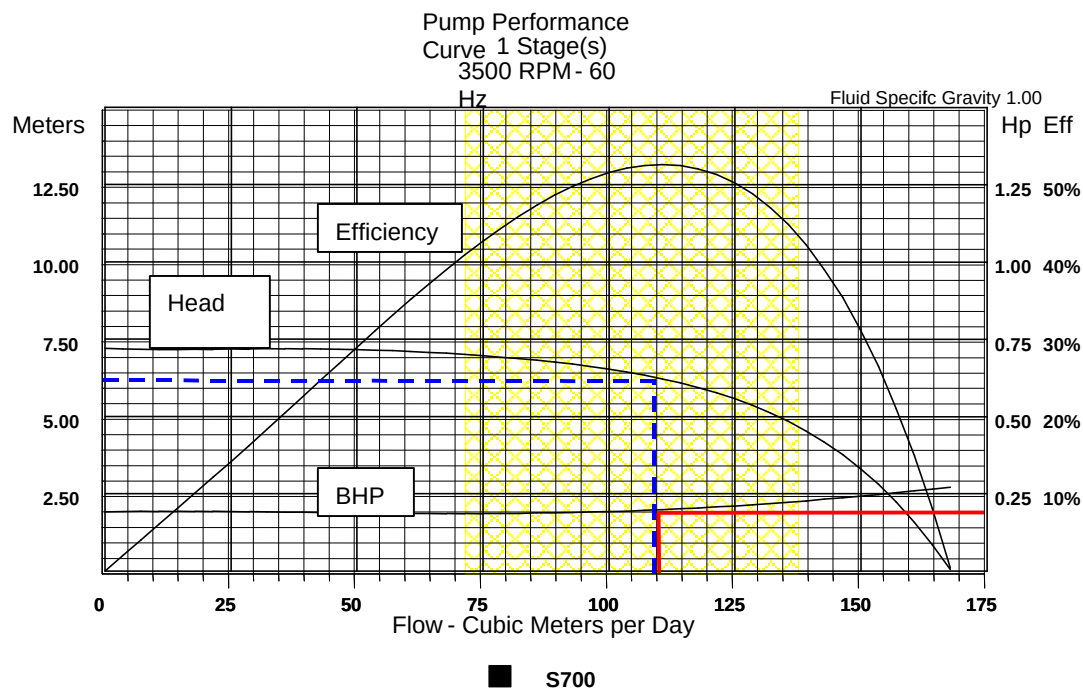


Figure 16: Altering Operating Point by Restricting Flow

In this instance, choking the well back to a rate of 110m³/d would shift the pump operating point to approximately the best efficiency point (BEP), i.e. the highest point on the efficiency curve. The pump would now be operating as recommended, and the likelihood of premature failure has been greatly reduced.

The second issue with this ESP system is the overloading of the motor. It can be seen in Figure 16 above that the power requirement per stage has been reduced from 0.25 BHP/stage to 0.2 BHP/stage. The pump power required is now:

$$\begin{aligned}\text{Total pump power required} &= \text{BHP/stage} \times \text{fluid SG} \times \text{No of stages} \\ &= 0.2 \times 0.92 \times 242 \\ &= 44.5 \text{ BHP.}\end{aligned}$$

$$\begin{aligned}\text{Total head} &= \text{head per stage} \times \text{No of stages} \\ &= 6.25 \times 242 \\ &= 1512.5 \text{ metres,}\end{aligned}$$

$$\text{from Figure 4 protector power} = 3.2 \text{ HP}$$

$$\begin{aligned}\text{Total power} &= \text{pump power} + \text{protector power} \\ &= 44.5 + 3.2 \\ &= 47.7 \text{ HP.}\end{aligned}$$

Converting this to kW gives $47.7 \times 0.746 = 35.6 \text{ kW}$.

This motor load of 35.6 kW is equivalent to 93% motor rating. Referring back to the motor curve in Figure 7, the current obtained for this loading is approx 26 A, or 90% nameplate current. The pump and motor would now be operating in their respective correct ranges, therefore mitigating the risk of premature failure. However, this has been achieved at the cost of reduced production; a correctly designed system would have allowed the higher production rate of 155m³/d with the ESP system functioning correctly.

2.2.4 Considerations when Inflow Performance Uncertainties Exist

Key Point:

- Where there is significant doubt as to inflow performance, it is preferable to undersize an ESP system as oppose to oversize.

Section 2.2.3 demonstrates actions which may be followed to alleviate problems encountered when well performance is stronger than predicted, and ultimately too strong for the ESP system deployed in the well (i.e. the pump is undersized). This results in a reduced production rate due to restricting the flowrate, however the pump will be operating correctly and risk of premature failure (due to upthrust at least) is minimised.

If the situation is considered where the inflow performance is overestimated and the well performance is too weak for the ESP installed in the well, the pump will then be oversized and will be operating in downthrust, again presenting a risk of premature pump failure due to operating outside the recommended range envelope. In the upthrust scenario, measures can be taken to move the operating point inside, or at least towards, the pump operating envelope by choking the well back as described above. However, in the case of an oversized pump operating in downthrust, the reservoir cannot supply enough fluids for the pump design criteria and there is no action which can be implemented to alleviate this problem (excepting stimulation treatments etc). Therefore in the oversized pump scenario the options are to either work the well over, replacing the pump with an alternative design, or accept that the operating conditions of the pump are likely to result in premature pump failure.

This demonstrates the requirement for accurate IPR data when planning an ESP system design. It also emphasizes that should any significant doubt exist as to the inflow performance of a well, it is prudent to undersize a pump as oppose to oversize due to the easier corrective action which may be taken if the IPR is not as predicted.

2.3 OVERSIZED PUMP

Example 3 illustrates the following important concepts in ESP operations:

- Increased power requirements between different ESP systems in the same well
- Transformers and their use in ESP systems
- Downthrust
- Well pump off and gas locking
- Underload protection

These issues are further discussed in the sections below.

2.3.1 Increased Power and Current Requirements

Key Point:

- The poorly designed system in example 3 has a surface current requirement (for the same applied surface voltage) of 237% of that of the efficient design in example 1, for the same well and flowrate. This illustrates the potential cost reductions that can be achieved in inefficient ESP systems.

Example 3 used the same well as example 1 with the intent of demonstrating the contrasting power requirements of two different ESP designs (one correctly designed and the other poorly designed) in the same well, delivering the same production rate.

Given that example 3 uses the same well as example 1, the lift requirements of the examples are identical. Therefore it is necessary to use 2 systems which deliver identical pump head values. This is achieved by selecting the appropriate number of stages for the M3000 pump in example 3 which delivers an equal amount of head as that generated by the S700 pump used in example 1. In section 1.7.3.1 in example 1, the total head generated at 110 m³/d was established as 1512.5 m. From the M3000 curve in Figure 9 above a head of 9.1 m /stage can be determined for the M3000 at 110 m³/d, therefore the number of stages required to generate the same head at the same flowrate is $1512.5/9.1 \approx 166$ stages.

The contrasting power requirements, for the same well at the same flowrate but with different ESP systems, can be seen in Table 10 below:

	Example 1	Example 3
Pump Power Required (kW)	35.6	82.2
Current Required (A)	27	30
Surface Voltage (V)	1172	2505

Table 10: Contrasting power requirements of Examples 1 and 3

The difference in pump power required is clearly apparent. However the difference in supplied electrical power is not so readily observed until the effects of the different surface voltages required, and subsequent transformer usage, have been evaluated.

2.3.2 Transformers

Between the ESP system and the electrical power supply (i.e. from national power system on land systems and from the rig supply on offshore systems) there will invariably be a transformer used to alter the voltage to that which is required by the ESP system. This is because ESP systems generally require voltages which are different from the available power supply voltage. Transformers which increase voltage are known as step up transformers, and transformers which decrease the voltage are known as step down. The current in the different transformer sections varies inversely as the voltage, i.e. in a step up transformer the output voltage will be greater than the input voltage but the output current will be decreased accordingly with regard to the input current. Transformers generally operate at relatively high efficiencies, between 97 and 99%, therefore transformer power losses can be ignored for most ESP system calculations.

In the event that current is measured prior to the transformer, the actual downhole current can be calculated from the following:

$$I_s = I_p \times (N_p/N_s)$$

Where I_p = primary current (surface supplied to transformer)

I_s = downhole current (after transformer)

N_p = number turns on the primary side of the transformer

N_s = number of turns on the secondary side of the transformer

The ratio N_p/N_s is also known as the transformation ratio, or turns ratio. Where the input and output voltages are known this can be calculated as follows:

$$(N_p/N_s) = (V_p/V_s)$$

Where V_p = primary voltage (input side of transformer)

V_s = secondary voltage (output, or ESP, side of transformer)

To obtain the different input electrical currents required in the respective examples, it is first required to calculate the transformation ratio of each transformer utilised. If we assume that the input voltage for each system is 380V (primary voltage), the transformation ratio of the transformer used in example 1 can be calculated as follows:

$$\begin{aligned} TR &= (N_p/N_s) = (V_p/V_s) \\ &= 380/1172 \\ &= 0.324. \end{aligned}$$

The surface current requirement can then be calculated:

$$\begin{aligned} I_p &= I_s/(N_p/N_s) \\ I_p &= 27/0.324 \\ I_p &\approx 83 \text{ A.} \end{aligned}$$

Similarly, the surface current requirement for example 3 can be calculated:

$$\begin{aligned} \text{TR} &= (N_p/N_s) = (V_p/V_s) \\ &= 380/2505 \\ &= 0.152 \end{aligned}$$

$$I_p = I_s/(N_p/N_s)$$

$$I_p = 30/0.152$$

$$I_p \approx 197 \text{ A.}$$

Therefore in example 3 the surface current requirement is 197 A, compared with 83 A for example 1, for identical electrical input supplies of 380 V. If the equipment used in example 3 was to be replaced with the equipment originally specified in example 1, an input current reduction of 57% would be observed (alternatively, the ESP system in example 3 uses 237% more current than the ESP system in example 1 for the same flowrate). Although the design in example 3 is extremely poor, this is a powerful demonstration of the current savings which may be achieved by appropriate equipment selection and correctly designed ESP systems. The principle of increased power consumption for the same work performed (and thus production rate obtained) is clearly demonstrated. This emphasises the need to optimise ESP designs in wells (considering matching the pump to the IPR and also the pump to the motor), particularly in regions where surface power requirements are an important economic factor.

2.3.3 Pump Operating Conditions - Downthrust

It is clear from the pump curve of Figure 11 that this pump is operating in downthrust. This is due to the pump capacity being greater than the production which the well inflow performance can deliver. As discussed in section 2.2.4 above, there is no action (with regard to pump operation) which can be taken to alleviate this problem. This will most likely result in premature pump failure.

2.3.4 Well Pump Off and Gas Locking

Key Point:

- Underload protection is vital in ESP systems to protect both the pump and motor from adverse conditions.

Section 2.3.4 is designed to demonstrate one of the effects (gas locking) that can occur in oversized ESP systems, specifically with the intent of demonstrating the requirement and necessity of underload protection in ESP systems.

As the pump rate in example 3 is greater than the well can deliver, this may result in the well being 'pumped off' and the pump gas locking. This may occur in an ESP whenever there is an excessive amount of free gas at the pump intake. Free gas (of a sufficient quantity) enters the intake stages and gas locks the pump; there is now no liquid entering or being pumped by the ESP.

The motor current requirements in this example were calculated as 30 A. In the situation where the well is being pumped off, the current would decrease gradually as the amount of

fluid available to the pump decreases (as the reservoir cannot sustain the pump rate and the intake pressure will be decreasing, increasing the proportion of free gas at the intake). At a certain point no fluid is available to the pump and the pump is gas locked.

It is more appropriate to analyse this situation from a logical perspective than a purely mathematical calculation of current. There are two main effects to consider:

1. A proportion of the fluid within the pump is now gas as oppose to liquid, which will result in a significant decrease in fluid SG
2. The flowrate through the pump has dropped to zero. From the pump curve in Figure 11 it can be seen that this results in a significant reduction in BHP/stage required.

Referring to the calculation for total power required by a pump:

Total pump power required = BHP/stage x fluid SG x No of stages

It can easily be appreciated that both of the effects listed above will contribute to a large reduction in pump power required, as two of the terms in the right hand side of the above equation will have decreased significantly. This in turn leads to a reduction in the current required by the motor. The SG will have changed dramatically when the fluid traversing the pump changed from oil to gas – depending on temperature and pressure at the pump depth, 'typical' gas SG figures are between 0.01 and 0.3 (these figures are considered in the average range – individual hydrocarbon reservoirs may vary outside this range). It can also be seen from the pump curve in Figure 11 that the power requirement per stage will have reduced from 0.7 to approximately 0.63 BHP/stage. If we assume that the fluid SG in the pump has an average value of 0.5, the power requirement becomes:

Total pump power required = $0.63 \times 0.5 \times 166$
= 52.3 BHP
= 39 kW.

If this value (which is equivalent to 36% motor rating) is then applied to the motor curve shown in Figure 10, an expected current of approximately 20 Amperes is obtained, which is equivalent to 67% of the normal operating current as calculated in section 1.9.3. The determination of absolute current requirement is irrelevant in this eventuality; the important principle is that the current requirement will be significantly lower than during normal operation.

Having established that in gas locking the current required in an ESP system is considerably lower than in normal operation, the significance of this must be analysed. The lowering of current in itself is not so much an issue in the motor, however the reduction of required current is generally a result of a greatly reduced flowrate through the pump. This has two key implications:

1. The pump will be operating outside it's recommended parameters therefore damage to the pump may occur
2. The motor cooling is completely dependent on liquid flow past the motor.

It is therefore apparent that both the pump and motor require protection from reduced current operation. In low or no flow conditions, motor burn out can occur in as little as one hour. Therefore preventing such an eventuality becomes crucial to prolonged pump and motor operation.

2.3.5 Underload Protection

Key Points:

- An underload trip setting of 80% normal operating current is generally accepted as a guidance point for underload setting, however this must be checked against BHP/stage at underload setting point on the pump curve prior to finalisation of the underload setting
- A correctly set underload current level can automatically protect a pump from operating in adverse conditions
- Correctly utilised underload protection can safeguard a motor from overheating to destruction should a situation of low or no flow occur at the pump
- The underload must be set above the idle amperage of the system – the idle amperage is the current the motor will draw when the pump it is attached to is performing no work. If the underload setting is below the idle amperage of the system, the unit may fall to the idle current when gas locking, for example, and eventually the motor would overheat (due to no cooling liquid flow past the motor) to the extent that insulation was broken down resulting in catastrophic motor failure
- The underload setting should also safeguard against deadheading, as described below.

The above example clearly demonstrates the requirement for automatic protection of ESP systems in the event of low currents. This is the purpose of underload protection. Underload protection is set to trip the pump when a certain low level of current is reached, thus protecting the pump from operating in extreme conditions and subsequent damage, and also protecting the motor from overheating due to low or no flow past the motor. Underload trips are generally set at a level of 80% of the normal operating current of the ESP system, however this requires to be manually checked in individual ESP systems which may vary in their requirements. For example, if a pump was operating at normal conditions at the lower end of the pump operating range (i.e. towards the left side of, but still within, the shaded area in the pump curve), simply setting the underload at 80% of the normal operating current may allow the pump to run in quite severe downthrust conditions yet the pump would not trip on underload. Therefore the permissible flowrate, and hence power requirements, of the pump must be considered, with respect to the pump curve, prior to setting the underload.

It is clear that underload protection is a vital part in the protection and correct operation of an ESP system, which can be summarised by the following:

2.3.6 Deadheading

Key Point:

- Underload settings should be checked on the pump curve to ensure that they protect against adverse flow conditions (e.g. severe downthrust) and deadheading.

When an ESP is operated against a closed (or blocked) surface valve, this is known as pump deadheading. Initial analysis of this situation might lead the operator to consider that this would result in an increased power requirement by the pump. The actual situation is just the opposite of this, however; once the void between the pump and the surface closure has been filled and sufficiently compressed, there is then no further liquid flow through the pump, although a constant head is being maintained above it. Generally, the lowest power requirement, and therefore lowest motor current requirement, occurs at zero flow conditions through the pump (as can be seen from a pump curve), therefore deadheading will result in a reduced current load.

Considering example 3 again, the power requirements can be determined for a deadheading situation (i.e. zero flow) from the pump curve as below in Figure 17.

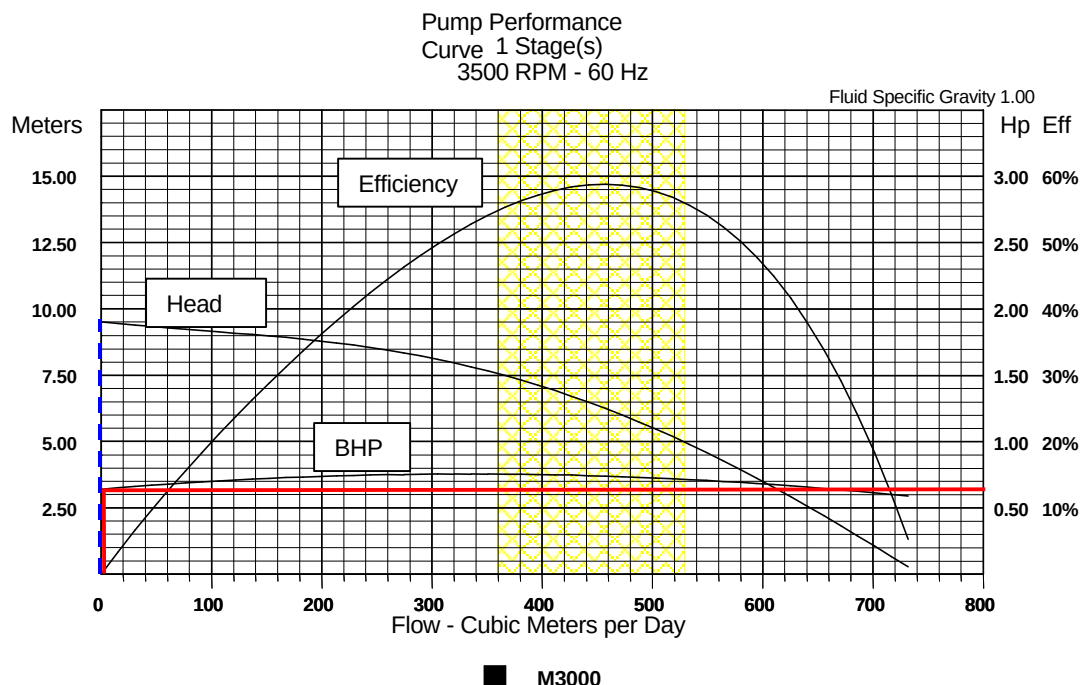


Figure 17: Determine power requirements and head for a deadheading system

$$\begin{aligned}
 \text{Total pump power required} &= \text{BHP/stage} \times \text{fluid SG} \times \text{No of stages} \\
 &= 0.64 \times 0.92 \times 166 \\
 &= 97.7 \text{ BHP.}
 \end{aligned}$$

$$\begin{aligned}
 \text{Total head} &= \text{head per stage} \times \text{No of stages} \\
 &= 9.5 \times 166 \\
 &= 1577 \text{ metres,}
 \end{aligned}$$

$$\text{from Figure 4 protector power} \approx 3.2 \text{ HP}$$

$$\begin{aligned}
 \text{Total power} &= \text{pump power} + \text{protector power} \\
 &= 97.7 + 3.2 \\
 &\approx 101 \text{ HP.}
 \end{aligned}$$

Converting this to kW gives $101 \times 0.746 \approx 75 \text{ kW}$, which is equivalent to 69% motor rating..

Applying this power requirement to the motor curve is demonstrated in Figure 18 below:

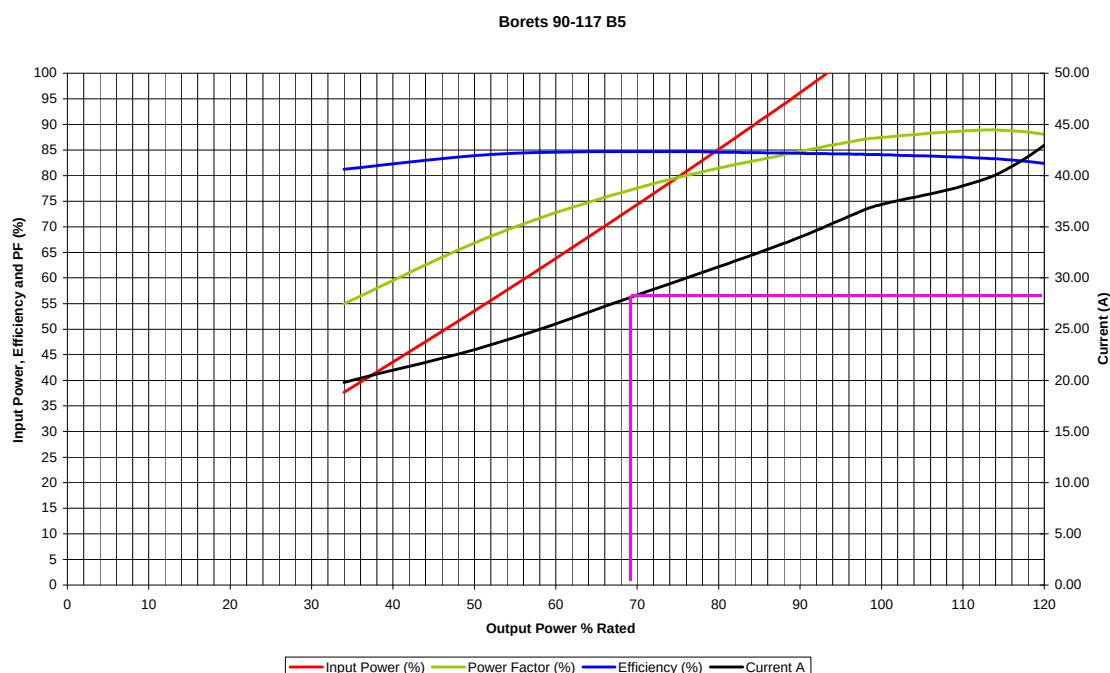


Figure 18: Determining current in a deadheading situation

A value of approximately 28 A is obtained, which is equivalent to 93% of the normal operating current of 30 A. Therefore to ensure that the underload setting would protect against deadheading occurring in example 3 the underload setting would have to be no lower than 93% of normal operating current. It should be considered that in this example the pump is running in quite severe downthrust; if the pump was operating close to the BEP the power per stage would be approximately 0.74 BHP (from Figure 17), which would result in an operating current of 31 A. Therefore in a correctly functioning pump of this size, the underload would have to be set to 90% of normal operating current to protect against deadheading. This illustrates that the pump curve should always be consulted before setting the underload trip level.

In the absence of motor curves, checking that the underload setting will prevent deadheading can be performed by checking the BHP per stage at zero flow conditions against the BHP per stage value during normal operation. For example, if the underload is set at 80% of normal operating current, the BHP/stage at deadhead (zero flow) conditions should be less than 80% of the BHP/stage at normal operation. This will generally ensure that the underload setting will protect against deadheading, providing that the current curve is relatively linear (this is usually the case).

In the event that a pump continually trips on underload (or overload), this indicates a problem with the system, and the system should be thoroughly checked before attempting repeated restarts.

2.4 VARIABLE SPEED DRIVE SYSTEMS

2.4.1 Effects of Variable Speed Systems on Pumps

Frequency changes result in specific effects on the behaviour of centrifugal pumps and induction motors. These have been introduced, along with the affinity laws, in section 1.10 and will be further examined here.

2.4.2 Affinity Laws

Key Points:

- The affinity law equations can be used to determine pump performance at differing frequencies
- To assess the effect of altering frequency on well fluid production a system nodal analysis must be performed

The parameters flowrate, frequency, head and BHP are related to each other in centrifugal pump systems by the following three relationships which collectively are known as the affinity laws:

Flowrate 1 = (Frequency 1/Frequency 2) x Flowrate 2

Head 1 = (Frequency 1/Frequency 2)² x Head 2

BHP 1 = (Frequency 1/Frequency 2)³ x BHP 2

where flowrate 1 is at frequency 1, flowrate 2 is at frequency 2, head 1 is at frequency 1, head 2 is at frequency 2, BHP 1 is at frequency 1 and BHP 2 is at frequency 2.

These laws allow the calculation of pump performance at frequencies other than that for which the pump curve is available. Care must be applied when using the affinity laws however in that the flow equation refers to the capacity of a centrifugal pump, i.e. this does not refer directly to an oil well application but instead to an unconstrained fluid source. More specifically this equation cannot be used to estimate the actual production change which will occur in an oil well as a result of a frequency change. This can only be achieved by performing a nodal analysis of the entire system which considers the lift characteristics of the well and pump along with the inflow capability of the reservoir, i.e. the productivity of the well.

It must also be noted that the affinity laws actually refer to frictionless incompressible flows, and are also strictly speaking only valid under constant efficiency conditions. However, they are routinely applied in the analysis of oilfield ESP systems. In certain engineering fields the exponent in the BHP equation may be varied between the value of 2 and 3, although the equations are generally applied as they appear above in oilfield applications.

2.4.3 Effects of Variable Speed Systems on Motors

Key Points:

In summation, a VSD system which varies voltage directly with frequency (as the vast majority of VSD systems do) results in a motor with the following characteristics:

- Output power rating varies directly with frequency
- Speed of rotation varies directly with frequency
- Voltage rating varies directly with frequency
- Current rating does not change with frequency
- Torque production does not change with frequency

Some of the effects of frequency changes on motors have previously been introduced in section 1.10 and will be expanded upon here.

The most effective manner in which to vary the speed of an induction motor is by varying the applied voltage in direct proportion to the frequency, i.e. maintain the V/f ratio as constant. The reasoning for this methodology is as follows. A parameter known as slip is defined as the relative velocity between the rotating magnetic field produced by the motor stator and the actual rotation of the rotor. For an induction motor to be efficient it must be operated at low values of slip, as slip is directly proportional to heat loss in the stator windings. For a given torque (the actual force produced by the motor), the magnitude of slip depends directly on the strength of the rotating magnetic field – the higher the magnetic field strength the lower the slip (for any given torque) and hence the higher the efficiency of the motor. Therefore it can be seen that it is highly desirable to maximise the magnetic field strength to its full rated value in the stator at all times. The magnetic field strength varies directly with applied voltage and inversely with applied frequency, therefore it can be seen that to maintain the magnetic field strength at a constant (maximum rated) value at different frequencies the applied voltage must be varied directly with frequency. This principle is central to variable speed induction motor, and hence ESP, systems. Additionally there is a limiting magnetic field strength which can be applied in an induction motor, otherwise the iron core will become saturated which results in excessive iron and copper losses in the motor, decreasing efficiency. This is why a large voltage value cannot simply be applied to the motor to increase power.

The above conditions results in a motor which delivers constant torque at all frequencies, apart from very low frequencies when stator electrical losses reduce available torque. This is normally compensated for by low speed voltage boosting, which provides the availability of constant torque even at low frequencies (additionally this technique also provides an advantage in starting in that full rated torque is available at low starting frequencies with low currents, unlike traditional fixed speed across the line starters which typically require starting currents in the order of 5 to 8 times that of current rating).

It must also be noted here that efficiency increases with frequency, as motor power output varies with frequency but electrical losses generally remain fairly constant. However, the efficiency variation over the frequency range employed in ESP systems is relatively minor.

2.4.4 Summary of Frequency Change Effects

This section is intended to list the changes which would be expected for a given frequency change in an ESP system. The effects have been listed for a frequency increase; the effects of a frequency decrease are simply the opposite of those listed here. A frequency increase in an ESP system will normally result in the following in an oil well:

- Voltage will increase (it is increased in proportion to frequency)
- Fluid production from the well will increase due to increased drawdown
- The BHP required per pump stage will increase
- Current will increase
- Head generated will increase
- Bottom hole flowing pressure will decrease due to increased head generated
- Well head pressure will most likely increase

2.4.5 Analysis of Frequency Change Effects on Well Performance

Although one of the affinity law equations calculates flow, this is only valid in the situation where there is unconstrained fluid available to the pump, i.e. the actual capacity of the pump as opposed to the flowrate obtained. To fully evaluate the effect which a frequency change has on production rate from a well requires a nodal analysis of the entire system, i.e. evaluation of vertical lift performance for the entire well at a range of production rates until the solution point with the inflow performance curve is found. This is a complex task which is generally performed with nodal analysis software. Superimposing the solution point on a pump tornado chart results in an illustration of a well's performance at different frequencies. A typical chart can be seen below in Figure 19.

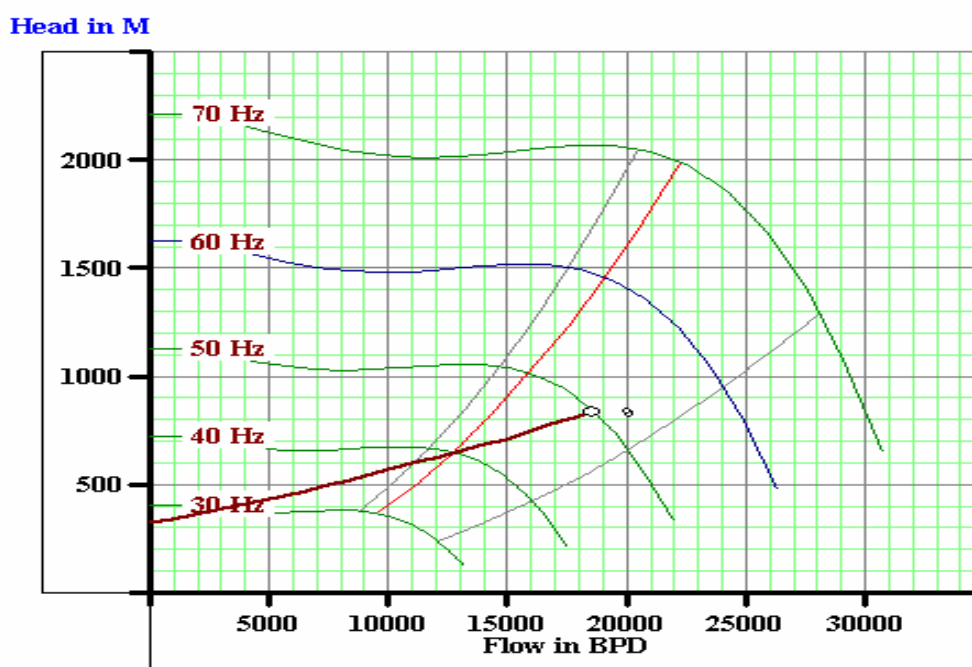


Figure 19: Well performance solution point on tornado chart

The shape of the curve obtained in a chart as seen above will obviously vary from well to well depending on several parameters including well geometry, productivity, reservoir pressure and ESP system utilised.

2.4.6 Potential VSD Benefits – Changing Well Conditions

Key Points:

- The increase in water cut (in the scenario outlined below) has resulted in an increase in motor loading which is too high for prolonged operation of the ESP
- Reducing the frequency is possible in a VSD system which will allow continued ESP operation

The ESP system in example 4 was operating close to the best efficiency point of the pump while the motor was operating at approx 99% rated power at the operating frequency of 53Hz. One potential advantage that a VSD can add to an ESP system is outlined below.

Consider the situation where the water production from the well in example 4 has risen significantly (and perhaps unexpectedly) to around 90%. The productivity of the well has remained relatively constant, therefore the production rate from the well has also remained constant (in reality the fluid rate will decrease due to increased hydrostatic fluid column pressure however for the sake of this example we will assume that the production rate at the pump has remained constant). As the well fluid SG will now have significantly changed, there will be a corresponding change in power requirement of the ESP system. If the water SG is 1.08, the new composite SG of the produced fluid will be:

$$(0.8 \times 0.1) + (1.08 \times 0.9) \\ = 1.05.$$

$$\begin{aligned} \text{Total pump power required} &= \text{BHP/stage} \times \text{fluid SG} \times \text{No of stages} \\ &= 0.69 \times 1.05 \times 155 \\ &\approx 112 \text{ BHP.} \end{aligned}$$

Adding the 2.9 HP required by the protector results in a new total system power requirement of approx 115 BHP. It was previously established that the motor is rated at 89.5 HP at a frequency of 53Hz, therefore the new power requirement is approx 128% of motor power at 53Hz. This is clearly not a desirable scenario; at 128% nameplate power the motor is going to burn out very quickly. In a fixed speed system this would undoubtedly result in a workover to change out the pump, motor or both. However, with a VSD in the system the frequency, or rotational speed, of the motor (and hence pump) can simply be altered to reduce the loading of the system.

As the frequency is reduced the power rating of the motor will decrease in proportion to the frequency. However due to the cubic nature of the BHP relationship in the affinity law equation, the pump power requirement will decrease significantly more than the motor rating decrease for a given frequency drop. Therefore at a certain lower operating frequency the pump power requirement will fall below that of the motor output, permitting pump and motor operation within the recommended envelope.

2.4.6.1 Discussion

Key Points:

- A VSD allows invaluable flexibility which can be used to maintain ESP operating conditions to within acceptable limits
- A VSD can be used to 'tune' an ESP in situ to changing well conditions.

It is clear that the flexibility that a VSD system provides would allow the well in this example to continue production when the water cut has risen to a level which would prohibit operation of a fixed speed system. In this example this has removed the requirement for a workover, which would otherwise be necessary in a fixed speed system.

It could be argued in this instance that the root cause of this problem was a fault of the original design and a fixed speed system should have been designed to cope with high water cuts, i.e. include a bigger motor in the design. However real life instances are not always so clear cut, for example reservoir simulation may have strongly suggested that no water breakthrough was expected until much later, at which time a workover was planned anyway, or perhaps there was a long lead time on a suitable motor etc. Field planning is subject to inherent uncertainties which manifest themselves in a variety of ways; however the important principle which is demonstrated in this example is the flexibility that a variable speed system provides. This equally applies to inflow performance issues, i.e. the flexibility that a VSD provides can be used to tune the ESP in situ to well performance, or optimise the operating point of the pump when inflow and well performance has not materialised as predicted.

2.4.7 Additional VSD Benefits and Considerations

It is not within the scope of this document to provide a detailed explanation of variable speed systems and their design; however a brief overview of certain other VSD characteristics will be presented here.

2.4.7.1 Design Considerations

It is imperative that all components in the system including the motor, surface power supply, pump shaft etc are selected considering the maximum frequency. If a lower frequency is used in the design process the ratings of the components may be exceeded.

2.4.7.2 Soft Start

The greatest period of stress in an ESP system is during start up. It is readily apparent that starting an ESP at a lower speed (frequency) and therefore power, will reduce the electrical and mechanical stresses experienced during startup. VSD systems are routinely used to 'soft start' ESP's at lower torque and current values than would be experienced by fixed speed systems, the speed can then be increased gradually and smoothly in an attempt to extend run life.

2.4.7.3 Electrical Advantages

A VSD can provide several electrical advantages including isolating the load (motor) from incoming switching and lightning transient voltages, eliminating frequency instabilities from generators and can provide some intelligent functions including automatic speed regulation, detection of backspin conditions and unit protection.

2.4.7.4 Operation at Lower Speed

In certain applications it may be advantageous to oversize the motor and run the system at low speeds, for example in highly abrasive conditions. This tends to occur in situations when pulling costs are high and the extended run lives more than compensate for the extra cost of the larger motor.